Grade 10 Term 1 Lesson 1 Practice Activity 1
Decision Alternatives, Events, Consequences, States, and Payoff Tables

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Evaluated criteria

C1. Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2. Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

1 Introduction

Problems:

1.1. Main Ideas and Concepts

1.2. Introductory General Problem (C1–C2)

1.3. From Supplied Payoffs to Calculated Models

1.4. Advanced General Problem (C1–C2)

1.5. Notation Reference and Practice Procedure

1.1 Main Ideas and Concepts

1. Start with the choice and its timing

A school enterprise offers a printing service for a school fair. Its student team must decide whether to offer a standard packet service or a colour packet service. The team must advertise one service before it knows how much demand there will be. First it chooses a service; later it learns whether demand is high or low.

The team is the *decision-maker*, and the two services are its *alternatives*. The uncertain process is how demand develops after advertising. High demand and low demand are the two possible *states of nature*: the possible outcomes of that process. In this simplified model, exactly one of those states occurs. The team controls its service choice, but it does not control the demand state.

A *consequence* tells us what a choice and a demand state mean for the enterprise. For example, offering standard packets under high demand could leave the team with money to buy supplies for its next project. A *payoff* measures the financial result in money. A consequence therefore explains what happens and why it matters; a payoff gives its numerical financial value.

Check. Is "high demand" the team's choice or an outcome of the uncertain process? *Suggested answer:* It is a possible outcome, or state. The team chooses the service before knowing the state.

2. Give the choices short names

Short names help us label tables without repeatedly writing long service descriptions. Call the standard packet service A_1, read "alternative one", and the colour packet service A_2, read "alternative two".

Symbol	Choice in the school enterprise
A_1	Offer the standard packet service
A_2	Offer the colour packet service

The small number below the letter is a *subscript*. It identifies a choice; it does not give a score or rank. For example, writing A_2 on a row means that the row is about the colour service.

Check. Does the 2 in A_2 mean that this service earns twice as much? *Suggested answer:* No. It is only the service's identifying number.

3. Give the possible demand outcomes short names

We also need short column labels for the possible demand outcomes. Call high demand S_1, read "state one", and low demand S_2, read "state two".

Symbol	Demand outcome
S_1	High demand
S_2	Low demand

Thus, if demand turns out to be low, we record S_2, whichever service the team chose.

Check. Can choosing A_1 guarantee S_1? *Suggested answer:* No. Choosing the standard service does not choose the demand state.

4. Match one choice with one state

To describe a possible consequence, we must say both which service is offered and which demand state occurs. We write (A_1, S_1) and read it as "standard packet service with high demand". This ordered pair names one possible combination; the alternative comes first and the state second.

There are four combinations to record:

Combination	Contextual reading
(A_1, S_1)	Standard service with high demand
(A_1, S_2)	Standard service with low demand
(A_2, S_1)	Colour service with high demand
(A_2, S_2)	Colour service with low demand

Before the choice and the demand outcome are known, our description must cover all four possibilities. After one service is chosen and one state occurs, only its particular combination is realised.

Check. If the team chooses colour service, should we erase the standard-service combinations from the original problem table? *Suggested answer:* No. The complete table records the consequences of both available choices under both states.

5. Attach a supplied monetary payoff

Suppose the enterprise is given a projected financial surplus of 20 USD for standard service with high demand. This means 20 USD remains available for its next project. We need a way to attach that number to the right combination. Write

$$P_{ay}(A_1, S_1) = 20 \text{ USD}.$$

Read the full statement as "the payoff from choosing standard service when high demand occurs is 20 US dollars". The name P_{ay} stands for payoff. Its two *arguments*, A_1 and S_1, identify the service and demand state. These parentheses identify inputs; they do not mean multiplication.

For this surplus measure, a positive payoff means money remains, zero means no surplus or loss, and a negative payoff means a loss that must be covered. For example, +20 USD can fund supplies, 0 USD leaves no money for the next

project, and −5 USD means the team must find 5 USD to cover a shortfall. These are examples of possible financial results, not extra components of the supplied 20 USD.

Check. Does $P_{ay}(A_1, S_1) = 20$ tell us how many packets were printed? *Suggested answer:* No. It supplies a monetary payoff, not an item count or a breakdown of that payoff.

6. Put each payoff in its place

Begin a payoff table with service names as rows and demand outcomes as columns. The supplied 20 USD belongs where the standard-service row meets the high-demand column. If the other supplied payoffs are 8 USD for standard service with low demand, 15 USD for colour service with high demand, and 6 USD for colour service with low demand, the completed table is:

Supplied payoffs (USD).

Service	High demand (S_1)	Low demand (S_2)
Standard (A_1)	20	8
Colour (A_2)	15	6

For example, the 6 in the second row and second column says $P_{ay}(A_2, S_2) = 6$ USD. Every cell links a consequence to a choice and a state. No calculation is needed when the payoff is supplied directly.

Check. Where would you look for the colour-service payoff under high demand? *Suggested answer:* Row A_2, column S_1: 15 USD.

The next general problem consolidates this same placement skill with generic choices and a different monetary scale. Its values are supplied independently; they are not the school enterprise's figures.

1.2 Introductory General Problem (C1–C2)

A decision-maker chooses between generic alternatives A_1 and A_2. After the choice, one uncertain event produces either state S_1 or state S_2. The payoffs, in thousand USD, are supplied directly:

$$P_{ay}(A_1, S_1) = 10, \quad P_{ay}(A_1, S_2) = 4, \quad P_{ay}(A_2, S_1) = 7, \quad P_{ay}(A_2, S_2) = 8.$$

Use the first argument to locate the alternative row and the second to locate the state column. The four combinations are

$$(A_1, S_1), \quad (A_1, S_2), \quad (A_2, S_1), \quad (A_2, S_2).$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: A_1	A_2: A_2
States of nature	S_1: S_1	S_2: S_2
Uncertain event	The uncertain event that produces either S_1 or S_2	
Consequences	The consequence and its payoff in thousand USD for each of the four listed combinations	



C2 - Building the payoff table.

Alternative	S_1	S_2
A_1	10	4
A_2	7	8

Because $P_{ay}(A_1, S_1)$ belongs to row A_1 and column S_1, 10 goes in the first payoff cell. Similarly, $P_{ay}(A_1, S_2) = 4$ goes in row A_1, column S_2; $P_{ay}(A_2, S_1) = 7$ goes in row A_2, column S_1; and $P_{ay}(A_2, S_2) = 8$ goes in row A_2, column S_2.

Check. What does the 7 in the completed table tell us? *Suggested answer:* Choosing A_2 when S_1 occurs gives a payoff of 7 thousand USD; it does not count items.



1.3 From Supplied Payoffs to Calculated Models

7. Extend the familiar table

The introductory problem has two alternatives and two states. To discuss any selected alternative without naming a specific row, write A_i, read "alternative with index i". To discuss any particular state, write S_j, read "state with index j". The letters i and j are identifying indices. For example, setting $i = 2$ gives $A_i = A_2$, and setting $j = 1$ gives $S_j = S_1$. In the introductory table, these substitutions locate $P_{ay}(A_2, S_1) = 7$ thousand USD.

We also need to say how many choices and states a larger problem contains. Let m be the *number of alternatives* and n the *number of states*. In the familiar two-by-two table, $m = 2$ and $n = 2$. A problem with three services and two demand outcomes would have $m = 3$ and $n = 2$:

$$A_1, A_2, A_3 \qquad \text{and} \qquad S_1, S_2.$$

In general, the lists are $A_1, \ldots, A_m$ and $S_1, \ldots, S_n$. The dots mean "continue the list". The index i runs from 1 to m, and j runs from 1 to n. An index identifies a row or column; a count tells us how many there are; a payoff is a financial value with units. These are different jobs.

Check. In the introductory problem, are $i = 2$, $m = 2$, and a payoff of 7 interchangeable? *Suggested answer:* No. They mean the selected row number, the total number of alternatives, and a payoff of 7 thousand USD.

Now we can write a general template. Each $P_{ay}(A_i, S_j)$, read "the payoff for alternative i under state j", still belongs at the intersection of that row and column.

Alternative	S_1	S_2	$\cdots$	S_n
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$\cdots$	$P_{ay}(A_1, S_n)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$\cdots$	$P_{ay}(A_2, S_n)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
A_m	$P_{ay}(A_m, S_1)$	$P_{ay}(A_m, S_2)$	$\cdots$	$P_{ay}(A_m, S_n)$

Check. If there are three alternatives and two states, how many payoff cells are needed? *Suggested answer:* Six: one for each of the 3×2 combinations.

8. Keep the table; change how its entries are obtained

Sometimes a problem supplies a financial result directly, as in the first general problem. Sometimes it supplies item counts, amounts earned per item, or fixed amounts instead. For example, "three packets, earning 2 USD each" requires a calculation before we can enter a monetary payoff. The rows and columns stay the same; the method for obtaining each entry changes.

We return to the school enterprise for the following *calculated-payoff variations*. Each variation states its financial arrangement explicitly. Its calculated payoffs replace, rather than reproduce, the earlier supplied figures. Unless stated otherwise, money is in USD and an item is one printed packet.

Check. May we put "3 packets" in a payoff table labelled USD? *Suggested answer:* No. We must first use the stated financial arrangement to find a monetary amount.

9. Record the quantity for each combination

A service may sell different numbers of packets under high and low demand. Colour packets may also attract different numbers of buyers from standard packets under the same demand conditions. We therefore record a quantity for each combination.

Suppose standard service sells 6 packets under high demand and 3 under low demand; colour service sells 4 and 2 packets, respectively. Write $q(A_1, S_2) = 3$ packets, read "the quantity for standard service under low demand is three packets". More generally, $q(A_i, S_j)$ means the quantity for the selected alternative and particular state.

Quantities (packets), not monetary payoffs.

Service	High demand (S_1)	Low demand (S_2)
Standard (A_1)	6	3
Colour (A_2)	4	2

For example, $q(A_2, S_1) = 4$ packets. The quantity can differ both across a row and down a column because it depends on the service and demand together. Use this quantity table throughout the school variations below.

Check. Does $q(A_1, S_2) = 3$ mean a payoff of 3 USD? *Suggested answer:* No. It means three packets; we still need the monetary terms.

10. Variation A: earnings per packet set by the service

Suppose each standard packet contributes 2 USD to the enterprise, and each colour packet contributes 3 USD. These are the stated earnings per packet after the relevant operating expenses; this variation includes no other payments. We need a rate determined by the service. Write $v(A_1) = 2$ USD per packet and $v(A_2) = 3$ USD per packet. Read $v(A_1)$ as "the variable payoff per item for standard service". In general, $v(A_i)$ is the alternative-based per-item payoff.

For a given service, this rate is the same in both demand states. Multiply its rate by the number of packets to obtain the monetary payoff:

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i).$$

Read this as "payoff equals quantity times the rate for the chosen alternative". For standard service under low demand, the compact worked example follows the input–calculation–final payoff sequence.

Symbol	Value	Meaning in this problem
$q(A_1, S_2)$	3 packets	Standard-service sales under low demand
$v(A_1)$	2 USD per packet	Standard-service earnings per packet

Symbolic calculation	Numerical calculation (USD)
$q(A_1, S_2) \times v(A_1)$	3×2
$= P_{ay}(A_1, S_2)$	$= 6$

Final payoff: 6 USD for standard service under low demand.

The units explain the multiplication: 3 packets times 2 USD per packet gives 6 USD. Under high demand the same service earns $6 \times 2 = 12$ USD. A constant rate across a row does not make the total payoff constant when quantities differ.

Check. Why is the standard-service rate 2 in both cells, while the payoffs are 12 and 6? *Suggested answer:* The rate is per packet; the quantities are six and three packets.

11. Variation B: earnings per packet set by demand

Consider a *separate one-component arrangement*: a sponsor pays for each packet, and these payments are the only payoff. The sponsor pays 1 USD per packet under high demand and 0.5 USD per packet under low demand, whichever service is chosen. These rates replace the service-based earnings in Variation A.

Write $v(S_1) = 1$ and $v(S_2) = 0.5$, both in USD per packet. Read $v(S_2)$ as "the variable payoff per item under low demand". In general, $v(S_j)$ is the state-based per-item payoff. Here the model is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j).$$

For standard service under low demand, collect the inputs and calculate this one cell:

Symbol	Value	Meaning in this problem
$q(A_1, S_2)$	3 packets	Standard-service sales under low demand
$v(S_2)$	0.5 USD per packet	Low-demand sponsor payment per packet

Standard service under low demand: one cell.

Symbolic calculation	Numerical calculation (USD)
$q(A_1, S_2) \times v(S_2)$	3×0.5
$= P_{ay}(A_1, S_2)$	$= 1.5$

Final payoff table (USD).

Alternative	S_2
A_1	1.5

Within a state column, the rate is the same for both services. The total need not be: colour service under low demand earns $2 \times 0.5 = 1$ USD because it sells only two packets.

Check. Does a shared column rate guarantee equal column payoffs? *Suggested answer:* No. Different services can sell different quantities in that state.

Extension AB: both per-packet earnings apply. Now suppose the enterprise receives both its service earnings from Variation A and the sponsor payments from Variation B for every packet. This is a new, two-component arrangement. Each packet generates two monetary contributions, so

$$\begin{aligned} P_{ay}(A_i, S_j) &= q(A_i, S_j)v(A_i) + q(A_i, S_j)v(S_j) \\ &= q(A_i, S_j)\big[v(A_i) + v(S_j)\big]. \end{aligned}$$

The first line counts the earnings from each source separately. The second adds the two rates first, then multiplies by their common quantity. Read the bracket as "service earnings per packet plus demand-based earnings per packet".

For standard service under low demand, both rates apply to the same three packets:

Symbol	Value	Meaning in this problem
$q(A_1, S_2)$	3 packets	Standard-service sales under low demand
$v(A_1)$	2 USD per packet	Service earnings for each packet
$v(S_2)$	0.5 USD per packet	Low-demand sponsor payment per packet

Standard service under low demand: one cell.

Symbolic calculation	Numerical calculation (USD)
$q(A_1, S_2)[v(A_1) + v(S_2)]$	$3[2 + 0.5]$
$= P_{ay}(A_1, S_2)$	$= 7.5$

Final payoff table (USD).

Alternative	S_2
A_1	7.5

This is also $3 \times 2 + 3 \times 0.5 = 7.5$ USD. Only compatible rates can be added: both here are USD per packet and apply to the same packets. A rate in USD per packet cannot be added directly to an amount in USD.

Check. Is $3(2 + 0.5)$ the same as $3 \times 2 + 0.5$? *Suggested answer:* No. Both rates apply to all three packets, so both are multiplied by 3.

12. Variation C: a fixed amount set by the service

Not every payment is earned once per packet. Suppose a school agreement provides a single payment of 5 USD for offering standard service or 4 USD for offering colour service at the fair, independent of how many packets are printed. These payments are *fixed*: within this model, they do not depend on quantity.

Write $f(A_1) = 5$ USD and $f(A_2) = 4$ USD. Read $f(A_1)$ as "the fixed payoff for standard service". In general, $f(A_i)$ is the alternative-based fixed payoff. In a *fixed-payment-only variation*, with no per-packet earnings, the model is

$$P_{ay}(A_i, S_j) = f(A_i).$$

For A_1 under either state, substituting $f(A_1) = 5$ USD gives a final payoff of 5 USD. The entire standard-service row is 5; the entire colour-service row is 4. No quantity multiplication is needed.

Extension AC: add the fixed agreement payment to Variation A. If the agreement payment is received in addition to the service's per-packet earnings, the model becomes

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i).$$

For standard service under low demand, use the packet count, service rate, and single agreement payment:

Symbol	Value	Meaning in this problem
$q(A_1, S_2)$	3 packets	Standard-service sales under low demand
$v(A_1)$	2 USD per packet	Service earnings for each packet
$f(A_1)$	5 USD	Single standard-service agreement payment

Standard service under low demand: one cell.

Symbolic calculation	Numerical calculation (USD)
$q(A_1, S_2) \times v(A_1) + f(A_1)$	$3 \times 2 + 5$
$= P_{ay}(A_1, S_2)$	$= 11$

Final payoff table (USD).

Alternative	S_2
A_1	11

Multiplying the fixed 5 USD by three would wrongly count a single agreement payment three times.

Check. If the number of packets doubles, does the agreement payment double? *Suggested answer:* No. The per-packet contribution doubles, but the fixed payment stays 5 USD for standard service.

13. Variation D: a fixed amount set by the state

A state can also trigger a single payment or cost. Suppose high demand brings a one-off school grant of 2 USD for the fair, while low demand brings a one-off unused-space charge of 1 USD. These amounts apply to either service and do not depend on the packet count.

Write $f(S_1) = 2$ USD and $f(S_2) = -1$ USD. Read $f(S_2)$ as "the fixed payoff under low demand". The negative sign records a cost that reduces the financial result. In general, $f(S_j)$ is the state-based fixed payoff. In a *state-payment-only variation*, with no other financial components,

$$P_{ay}(A_i, S_j) = f(S_j).$$

For either service under low demand, the input $f(S_2) = -1$ USD gives a final payoff of -1 USD. Under high demand the final payoff is 2 USD. Thus each column is constant in this one-component case. "Fixed" means independent of quantity, not identical across states.

Compare the two low-demand terms: $v(S_2) = 0.5$ USD *per packet* in Variation B, but $f(S_2) = -1$ USD *once for the fair* in Variation D. Three packets would generate $3 \times 0.5 = 1.5$ USD from the first term, while the second term would still be -1 USD.

Check. Should three packets make the unused-space charge -3 USD? *Suggested answer:* No. It is a single 1 USD charge, so its contribution is -1 USD.

14. Assemble the complete model

Complete extension ABCD. Suppose all four arrangements now apply together: service earnings and sponsor payments for each packet, plus the service agreement payment and the state-dependent grant or charge. These are the same four sources already explained, now combined in one financial result:

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i) + f(S_j).$$

Read this as "quantity times the sum of the two per-item rates, plus the alternative's fixed amount, plus the state's fixed amount".

For standard service under low demand, collect the familiar inputs first:

Symbol	Value	Meaning in this problem
$q(A_1, S_2)$	3 packets	Standard-service sales under low demand
$v(A_1)$	2 USD per packet	Service earnings for each packet
$v(S_2)$	0.5 USD per packet	Sponsor payment for each packet
$f(A_1)$	5 USD	Single standard-service agreement payment
$f(S_2)$	-1 USD	Single low-demand unused-space charge

Standard service under low demand: one cell.

Symbolic calculation	Numerical calculation (USD)
$q(A_1, S_2)[v(A_1) + v(S_2)]$	$3[2 + 0.5]$
$+ f(A_1) + f(S_2)$	$+ 5 - 1$
$= P_{ay}(A_1, S_2)$	$= 11.5$

Final payoff table (USD).

Alternative	S_2
A_1	11.5

The final payoff is 11.5 USD: money available for the next project after these contributions and the charge are included. This differs from Variation A's 6 USD because the financial arrangement now includes three additional components.

The four additive payoff components are $q(A_i, S_j)v(A_i)$, $q(A_i, S_j)v(S_j)$, $f(A_i)$, and $f(S_j)$. The quantity $q(A_i, S_j)$ multiplies the two variable components; it is not a fifth additive payoff component. Each of the four components has monetary units after any multiplication.

An absent component is set to zero. For example, setting $v(S_j) = 0$, $f(A_i) = 0$, and $f(S_j) = 0$ leaves the one-component formula $P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i)$. This does not assume that every real arrangement includes four payments; it is a way to describe simpler models using the same structure.

Check. Why do five input types produce only four payoff components? *Suggested answer:* Quantity is a multiplier used in two components, not a separate amount of money to add.

The practice sections follow this progression. A direct-payoff model supplies each cell. One-component models use just one of the four monetary components. Two-component models add two, for example $q(A_i, S_j)v(A_i) + f(A_i)$. Three-component models add three, for example $q(A_i, S_j)[v(A_i) + v(S_j)] + f(A_i)$. Complete four-component models include all four. The number of components counts monetary contributions, not symbols or table cells.

An argument also tells us which input determines a component. A rate or fixed amount with argument A_i is determined by the alternative and is reused across its row. One with argument S_j is determined by the state and is reused down its column. A quantity with both arguments is specified for each combination. Values need not actually differ: two services may happen to have equal rates, even though each rate is specified by service.

Check. If both services earn 2 USD per packet, can the rates still be alternative-based? *Suggested answer:* Yes. The arrangement specifies the rate for each service; the two values happen to match.

15. Bring the familiar steps together

The Advanced General Problem below is an independent example in thousand USD. It uses the same quantities, per-item rates, and fixed-amount notation, with new supplied values. Start with its *Symbol | Value | Meaning in this problem* table. Then match each symbolic cell to the numerical cell beside it: alternatives and states remain in the same order. Finally, copy the evaluated results into a clean payoff table. In every numerical calculation cell, the arithmetic expression is on one line and the result is directly below it.

Each number has a familiar job. Use the full worked tables to trace all four cells in the same way.

1.4 Advanced General Problem (C1–C2)

A decision-maker chooses A_1 or A_2 before an uncertain event produces S_1 or S_2. All monetary values below are in thousand USD, with variable values stated per item:

$$v(A_1) = 2, \quad v(A_2) = 3, \qquad f(A_1) = 5, \quad f(A_2) = 6,$$

$$v(S_1) = 1, \quad v(S_2) = 2, \qquad f(S_1) = 1, \quad f(S_2) = 2.$$

The alternative–state quantities are:

Alternative	S_1	S_2
A_1	$q(A_1, S_1) = 2$	$q(A_1, S_2) = 3$
A_2	$q(A_2, S_1) = 4$	$q(A_2, S_2) = 5$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: A_1	A_2: A_2
States of nature	S_1: S_1	S_2: S_2
Uncertain event	The uncertain event that produces either S_1 or S_2	
Consequences	The total payoff in thousand USD for each pair (A_i, S_j)	

C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Generic alternative 1.
A_2	—	Alternative: Generic alternative 2.
S_1	—	State: Generic state 1.
S_2	—	State: Generic state 2.
$q(A_1, S_1)$	2 items	Quantity for Generic alternative 1 under Generic state 1.
$q(A_1, S_2)$	3 items	Quantity for Generic alternative 1 under Generic state 2.
$q(A_2, S_1)$	4 items	Quantity for Generic alternative 2 under Generic state 1.
$q(A_2, S_2)$	5 items	Quantity for Generic alternative 2 under Generic state 2.
$v(A_1)$	2 thousand USD per item	Alternative-based earnings for Generic alternative 1.
$v(A_2)$	3 thousand USD per item	Alternative-based earnings for Generic alternative 2.
$v(S_1)$	1 thousand USD per item	State-based earnings for Generic state 1.
$v(S_2)$	2 thousand USD per item	State-based earnings for Generic state 2.
$f(A_1)$	5 thousand USD	Alternative-based fixed income for Generic alternative 1.
$f(A_2)$	6 thousand USD	Alternative-based fixed income for Generic alternative 2.
$f(S_1)$	1 thousand USD	State-based fixed income for Generic state 1.
$f(S_2)$	2 thousand USD	State-based fixed income for Generic state 2.

For example, $q(A_1, S_1) = 2$ counts items; $v(A_1) = 2$ and $v(S_1) = 1$ are thousand USD per item; $f(A_1) = 5$ and $f(S_1) = 1$ are thousand USD counted once. Thus the first cell is $2(2 + 1) + 5 + 1 = 12$ thousand USD.
The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i) + f(S_j).$$

Advanced General Problem: payoff calculations.

All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\big[v(A_1)+v(S_1)\big]$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\big[v(A_1)+v(S_2)\big]$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\big[v(A_2)+v(S_1)\big]$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\big[v(A_2)+v(S_2)\big]$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$2\big[2+1\big]$ $+5+1$ $=12$	$3\big[2+2\big]$ $+5+2$ $=19$
A_2	$4\big[3+1\big]$ $+6+1$ $=23$	$5\big[3+2\big]$ $+6+2$ $=33$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	12	19
A_2	23	33

1.5 Notation Reference and Practice Procedure

The reference below collects notation already used. Always attach the units stated in the particular problem; the school examples used USD, while the general problems used thousand USD.

Symbol	Meaning and units
A_i, S_j	The selected alternative and particular state; labels, not money
i, j	Indices identifying a row and a column
m, n	Counts of alternatives and states
(A_i, S_j)	One alternative–state combination
$q(A_i, S_j)$	Quantity for that combination, in items
$v(A_i)$, $v(S_j)$	Alternative-based and state-based rates, in money per item
$f(A_i)$, $f(S_j)$	Alternative-based and state-based fixed amounts, in money
$Pay(A_i, S_j)$	Total payoff for that combination, in money

Procedure for the practice problems.

1. Identify who decides, the available choices, and the uncertain process. State what is chosen first and which state is learned later. Describe what the consequences mean economically (C1).
2. Name the alternatives and states, and set up rows and columns for every combination (C2).
3. If payoffs are supplied, place them directly in their cells. If calculation is needed, first build a *Symbol* / *Value* / *Meaning in this problem* input table with units and signs. Identify which financial components actually apply.
4. For calculated models, build matching symbolic and numerical tables side by side. Substitute the inputs for each combination. Keep the arithmetic expression on one line and its result below it.
5. Collect only the evaluated monetary values in a final payoff table. Check row and column placement, signs, units, and completeness. The task ends with an organised description of the possible financial results.

Final checks for understanding.

1. A cell lists a quantity of 4 items. Is 4 ready to enter in a payoff table measured in thousand USD? *Suggested answer:* No. Quantity counts items; use the applicable monetary rates and fixed amounts to calculate the payoff.
2. A payment is 3 USD per item and a grant is 5 USD once. For two items, should both be multiplied by two? *Suggested answer:* No. The total is $2 \times 3 + 5 = 11$ USD; the grant is counted once.
3. Which input is reused across a row: $v(A_i)$ or $v(S_j)$? Does that make every payoff in the row equal? *Suggested answer:* $v(A_i)$ is reused across the row. Payoffs can still differ because quantity or state-based components can differ. The state-based rate $v(S_j)$ is reused down a column.



2 Direct Payoff Models

Problems:

2. Local Café Investment (C1–C2)
2. Green Energy Production Mix (C1–C2)
2. Global Shipping Network Design (C1–C2)

Local Café Investment (C1–C2)

A small café is deciding whether to introduce a line of artisan desserts or keep its current menu. The payoff represents estimated daily profit from café sales. The projections combine recent sales records with expected ingredient, staffing, and preparation costs.

The result depends on customer traffic, which may be high or low. If the café launches the dessert line, the projected profit is 42 thousand USD under high traffic and 8 thousand USD under low traffic. The large difference reflects the additional sales possible when demand is strong and the risk of unsold ingredients when demand is weak.

If the café keeps its current menu, the projected profit is 28 thousand USD under high traffic and 20 thousand USD under low traffic. This alternative offers less growth potential but produces more stable outcomes because it avoids most of the new operating costs.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Launch dessert line	A_2: Keep current menu
States of nature	S_1: High traffic	S_2: Low traffic
Uncertain event	Actual foot traffic after the decision	
Consequences	Profit in thousand USD for each alternative–state pair	



C2 - Building the payoff table.

Alternative	High traffic	Low traffic
Launch dessert line	42	8
Keep menu	28	20



Green Energy Production Mix (C1–C2)

A renewable-energy company must choose among building wind turbines, developing solar farms, or installing a balanced hybrid system. The payoff represents projected annual profit from electricity generation and sales. The estimates are based on expected production capacity, electricity prices, and operating and maintenance costs.

Profit depends on whether conditions are windy, sunny, or cloudy. For the wind-turbine strategy, the projected profits are 90 thousand USD under windy conditions, 45 thousand USD under sunny conditions, and 30 thousand USD under cloudy conditions.

For the solar-farm strategy, the projected profits are 35 thousand USD under windy conditions, 95 thousand USD under sunny conditions, and 25 thousand USD under cloudy conditions. The hybrid system is expected to produce 70 thousand USD under windy conditions, 65 thousand USD under sunny conditions, and 60 thousand USD under cloudy conditions.

The specialized wind and solar systems perform best under favorable weather but decline sharply under less suitable conditions. The hybrid system does not achieve the highest possible payoff, but its outcomes are more stable because it combines two energy sources.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Wind turbines	A_2: Solar farms	A_3: Hybrid system
States of nature	S_1: Windy	S_2: Sunny	S_3: Cloudy
Uncertain event	Weather outcome affecting generation		
Consequences	Profit in thousand USD for each strategy–weather combination		



C2 - Building the payoff table.

Strategy	Windy	Sunny	Cloudy
Wind turbines	90	45	30
Solar farms	35	95	25
Hybrid system	70	65	60



Global Shipping Network Design (C1–C2)

A multinational logistics firm must choose between a centralized mega-hub network and a regional multi-hub network for its long-term freight contracts. The payoff represents projected profit after transportation, storage, coordination, and delay costs. Management developed the estimates from previous contracts, expected freight volumes, and simulated disruption scenarios.

The firm considers four possible trade conditions: a trade boom, stable trade, moderate disruptions, and severe disruptions. Under the centralized network, projected profits are 220 thousand USD during a trade boom, 140 thousand USD during stable trade, and 40 thousand USD during moderate disruptions; severe disruptions would produce a loss of 30 thousand USD. This design benefits from economies of scale when trade is strong but becomes vulnerable when disruptions affect the main hub.

Under the regional network, projected profits are 180, 160, 110, and 60 thousand USD under the same four conditions. Operating several hubs limits the highest possible profit, but the network remains more resilient because freight can be redirected when disruptions occur.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description			
Alternatives	A_1: Centralized mega-hub	A_2: Regional multi-hub		
States of nature	S_1: Boom	S_2: Stable	S_3: Moderate disruptions	S_4: Severe disruptions
Uncertain event	Trade conditions realized during the planning horizon			
Consequences	Profit in thousand USD for each network–state pair			



C2 - Building the payoff table.

Network	Boom	Stable	Moderate disruptions	Severe disruptions
Centralized mega-hub	220	140	40	-30
Regional multi-hub	180	160	110	60



3 One-Component Models

Problems:

3.1. Community Bakery Lunch Boxes (C1–C2)

3.1. Rural Internet Subscriptions (C1–C2)

3.2. Coffee Export Routes (C1–C2)

3.2. Lake Ferry Tourism (C1–C2)

3.3. School Meal Service Contracts (C1–C2)

3.3. Mobile Clinic Service Contracts (C1–C2)

3.4. Neighborhood Shelter Support (C1–C2)

3.4. Community Hall Emergency Rentals (C1–C2)

3.1 Alternative-Based Variable

Problems:

3.1. Community Bakery Lunch Boxes (C1–C2)

3.1. Rural Internet Subscriptions (C1–C2)

Community Bakery Lunch Boxes (C1–C2)

A community bakery is bidding for office lunch-box contracts and must choose between standard and premium packaging. The payoff depends on the profit earned per lunch box and the number of boxes sold. The bakery estimates demand using recent office orders, client inquiries, and projected packaging costs.

Standard packaging generates a profit of 0.04 thousand USD per lunch box. The bakery expects to sell 900 boxes under high office demand and 550 boxes under low demand. Premium packaging generates a higher profit of 0.07 thousand USD per box, but its higher price is expected to reduce sales to 650 boxes under high demand and 250 boxes under low demand.

The premium option earns more per sale but is more sensitive to weak demand, while standard packaging attracts a larger number of customers in both demand states.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Standard packaging	A_2: Premium packaging
States of nature	S_1: High demand	S_2: Low demand
Uncertain event	Actual number of lunch-box contracts realized	
Consequences	Profit in thousand USD, based on boxes × profit per box	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Standard packaging.
A_2	—	Alternative: Premium packaging.
S_1	—	State: High demand.
S_2	—	State: Low demand.
$q(A_1, S_1)$	900 lunch boxes	Quantity for Standard packaging under High demand.
$q(A_1, S_2)$	550 lunch boxes	Quantity for Standard packaging under Low demand.
$q(A_2, S_1)$	650 lunch boxes	Quantity for Premium packaging under High demand.
$q(A_2, S_2)$	250 lunch boxes	Quantity for Premium packaging under Low demand.
$v(A_1)$	0.04 thousand USD per lunch box	Packaging profit for Standard packaging.
$v(A_2)$	0.07 thousand USD per lunch box	Packaging profit for Premium packaging.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i).$$

Community Bakery Lunch Boxes: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times v(A_1)$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2) \times v(A_1)$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1) \times v(A_2)$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2) \times v(A_2)$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	900×0.04 $= 36$	550×0.04 $= 22$
A_2	650×0.07 $= 45.5$	250×0.07 $= 17.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	36	22
A_2	45.5	17.5



Rural Internet Subscriptions (C1–C2)

A rural internet provider must decide whether to promote its Basic, Plus, or Pro subscription plan. The payoff depends on the profit earned per subscription and the number of customers who enroll. The estimates are based on previous enrollment patterns, local market surveys, and projected service and installation costs.

The Basic plan generates a profit of 0.03 thousand USD per subscription and is expected to attract 1400 customers under high enrollment and 900 under low enrollment. The Plus plan generates 0.05 thousand USD per subscription, with projected sales of 1000 under high enrollment and 700 under low enrollment.

The Pro plan produces the highest profit per subscription, at 0.08 thousand USD, but is expected to attract only 600 customers under high enrollment and 300 under low enrollment. The pattern reflects the trade-off between higher profit per customer and lower demand for more expensive plans.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Basic plan	A_2: Plus plan	A_3: Pro plan
States of nature	S_1: High enrollment	S_2: Low enrollment	
Uncertain event	Actual subscription uptake after plan selection		
Consequences	Profit in thousand USD based on subscriptions × profit per subscription		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Basic plan.
A_2	—	Alternative: Plus plan.
A_3	—	Alternative: Pro plan.
S_1	—	State: High enrollment.
S_2	—	State: Low enrollment.
$q(A_1, S_1)$	1400 subscriptions	Quantity for Basic plan under High enrollment.
$q(A_1, S_2)$	900 subscriptions	Quantity for Basic plan under Low enrollment.
$q(A_2, S_1)$	1000 subscriptions	Quantity for Plus plan under High enrollment.
$q(A_2, S_2)$	700 subscriptions	Quantity for Plus plan under Low enrollment.
$q(A_3, S_1)$	600 subscriptions	Quantity for Pro plan under High enrollment.
$q(A_3, S_2)$	300 subscriptions	Quantity for Pro plan under Low enrollment.
$v(A_1)$	0.03 thousand USD per subscription	Plan profit for Basic plan.
$v(A_2)$	0.05 thousand USD per subscription	Plan profit for Plus plan.
$v(A_3)$	0.08 thousand USD per subscription	Plan profit for Pro plan.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i).$$

Rural Internet Subscriptions: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times v(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times v(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(A_2)$ $=P_{ay}(A_2,S_2)$
A_3	$q(A_3,S_1)\times v(A_3)$ $=P_{ay}(A_3,S_1)$	$q(A_3,S_2)\times v(A_3)$ $=P_{ay}(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	1400×0.03 $=42$	900×0.03 $=27$
A_2	1000×0.05 $=50$	700×0.05 $=35$
A_3	600×0.08 $=48$	300×0.08 $=24$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	42	27
A_2	50	35
A_3	48	24

3.2 State-Based Variable

Problems:

3.2. Coffee Export Routes (C1–C2)
3.2. Lake Ferry Tourism (C1–C2)

Coffee Export Routes (C1–C2)

A coffee exporter must choose whether to ship its products through Port A or Port B. The payoff depends on the profit earned per shipment and the number of shipments delivered. Management developed the estimates from previous export volumes, transportation costs, port capacity, and exchange-rate scenarios.

The exchange rate may be favorable or unfavorable. A favorable rate produces a profit of 6 thousand USD per shipment, while an unfavorable rate reduces the profit to 2 thousand USD per shipment.

Through Port A, the exporter expects to deliver 40 shipments under a favorable exchange rate and 28 under an unfavorable rate. Through Port B, it expects to deliver 32 shipments under a favorable rate and 30 under an unfavorable rate. Port A offers greater capacity in favorable conditions, while Port B maintains more stable shipment volumes across both states.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Port A route	A_2: Port B route
States of nature	S_1: Favorable rate	S_2: Unfavorable rate
Uncertain event	Exchange-rate level that affects profit per shipment	
Consequences	Profit in thousand USD based on shipments × profit per shipment	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Port A route.
A_2	—	Alternative: Port B route.
S_1	—	State: Favorable rate.
S_2	—	State: Unfavorable rate.
$q(A_1, S_1)$	40 shipments	Quantity for Port A route under Favorable rate.
$q(A_1, S_2)$	28 shipments	Quantity for Port A route under Unfavorable rate.
$q(A_2, S_1)$	32 shipments	Quantity for Port B route under Favorable rate.
$q(A_2, S_2)$	30 shipments	Quantity for Port B route under Unfavorable rate.
$v(S_1)$	6 thousand USD per shipment	Exchange-rate profit for Favorable rate.
$v(S_2)$	2 thousand USD per shipment	Exchange-rate profit for Unfavorable rate.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j).$$

Coffee Export Routes: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times v(S_1)$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2) \times v(S_2)$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1) \times v(S_1)$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2) \times v(S_2)$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	40×6 $= 240$	28×2 $= 56$
A_2	32×6 $= 192$	30×2 $= 60$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	240	56
A_2	192	60



Lake Ferry Tourism (C1–C2)

A lake tourism operator must decide whether to use small ferries or large ferries for sightseeing tours. The payoff depends on the profit earned per ticket and the number of tickets sold. The estimates are based on previous passenger demand, seasonal ticket prices, and projected operating costs.

The operator considers peak season, normal season, and rainy season. Profit per ticket is 0.05 thousand USD during peak season, 0.04 thousand USD during normal season, and 0.03 thousand USD during rainy season.

Small ferries are expected to sell 2200 tickets in peak season, 1600 in normal season, and 1000 in rainy season. Large ferries are expected to sell 2600, 1800, and 800 tickets under the same respective conditions. Large ferries can serve more passengers when demand is strong, but their ticket sales fall below those of small ferries during the rainy season.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Small ferries	A_2: Large ferries	
States of nature	S_1: Peak	S_2: Normal	S_3: Rainy
Uncertain event	Tourism season and ticket price level		
Consequences	Profit in thousand USD based on tickets $\times$ profit per ticket		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Small ferries.
A_2	—	Alternative: Large ferries.
S_1	—	State: Peak.
S_2	—	State: Normal.
S_3	—	State: Rainy.
$q(A_1, S_1)$	2200 tickets	Quantity for Small ferries under Peak.
$q(A_1, S_2)$	1600 tickets	Quantity for Small ferries under Normal.
$q(A_1, S_3)$	1000 tickets	Quantity for Small ferries under Rainy.
$q(A_2, S_1)$	2600 tickets	Quantity for Large ferries under Peak.
$q(A_2, S_2)$	1800 tickets	Quantity for Large ferries under Normal.
$q(A_2, S_3)$	800 tickets	Quantity for Large ferries under Rainy.
$v(S_1)$	0.05 thousand USD per ticket	Seasonal ticket profit for Peak.
$v(S_2)$	0.04 thousand USD per ticket	Seasonal ticket profit for Normal.
$v(S_3)$	0.03 thousand USD per ticket	Seasonal ticket profit for Rainy.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j).$$

Lake Ferry Tourism: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$q(A_1,S_1)\times v(S_1)$ $= P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(S_2)$ $= P_{ay}(A_1,S_2)$	$q(A_1,S_3)\times v(S_3)$ $= P_{ay}(A_1,S_3)$
A_2	$q(A_2,S_1)\times v(S_1)$ $= P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(S_2)$ $= P_{ay}(A_2,S_2)$	$q(A_2,S_3)\times v(S_3)$ $= P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	2200×0.05 $=110$	1600×0.04 $=64$	1000×0.03 $=30$
A_2	2600×0.05 $=130$	1800×0.04 $=72$	800×0.03 $=24$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	110	64	30
A_2	130	72	24

3.3 Alternative-Based Fixed

Problems:

3.3. School Meal Service Contracts (C1–C2)
3.3. Mobile Clinic Service Contracts (C1–C2)

School Meal Service Contracts (C1–C2)

A school cafeteria provider must choose a basic or extended meal contract for a short academic program. After projected food, staffing, and operating costs, the agreed service fee leaves 18 thousand USD under the basic contract and 32 thousand USD under the extended contract.

Participation may be regular or high, but the contracts were negotiated in advance, so their financial results do not change with participation.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Basic meal	A_2: Extended meal
States of nature	S_1: Regular participation	S_2: High participation
Uncertain event	Participation level observed after the contract choice	
Consequences	Fixed payoff in thousand USD tied only to the selected contract	



C2 - Building the payoff table.

Alternative	Regular participation	High participation
Basic meal	18	18
Extended meal	32	32



Mobile Clinic Service Contracts (C1–C2)

A health nonprofit must select a basic outreach, extended outreach, or full regional contract for mobile clinics. After expected staffing, transportation, equipment, and operating costs, contracted funding leaves 24 thousand USD for basic outreach, 38 thousand USD for extended outreach, and 52 thousand USD for full regional service.

Community need may be regular or high, but the funding was agreed before need is observed, so the financial result for each contract remains unchanged.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Basic outreach	A_2: Extended outreach	A_3: Full regional
States of nature	S_1: Regular need	S_2: High need	
Uncertain event	Community-need level observed after the contract choice		
Consequences	Fixed payoff in thousand USD tied only to the selected contract		

C2 - Building the payoff table. Calculation table in thousand USD:

Payoff = contract fixed payoff

Alternative	Regular need	High need
Basic outreach	24	24
Extended outreach	38	38
Full regional	52	52

Alternative	Regular need	High need
Basic outreach	24	24
Extended outreach	38	38
Full regional	52	52

3.4 State-Based Fixed

Problems:

3.4. Neighborhood Shelter Support (C1–C2)

3.4. Community Hall Emergency Rentals (C1–C2)

Neighborhood Shelter Support (C1–C2)

A municipal office must decide whether to open a school gym or a sports center as a temporary shelter. Its emergency budget provides 12 thousand USD during a minor emergency and 30 thousand USD during a major emergency, based on expected shelter operating requirements.

The same support is available for either facility because funding depends on emergency severity, and the larger response receives more support.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: School gym	A_2: Sports center
States of nature	S_1: Minor emergency	S_2: Major emergency
Uncertain event	Emergency level realized after the shelter decision	
Consequences	Fixed payoff in thousand USD tied only to the emergency state	



C2 - Building the payoff table.

Alternative	Minor emergency	Major emergency
School gym	12	30
Sports center	12	30



Community Hall Emergency Rentals (C1–C2)

A city council must decide whether to reserve one central hall or two neighborhood halls for emergency use. Budget forecasts provide 18 thousand USD of facility-rental support during a mild emergency, 35 thousand USD during a serious emergency, and 50 thousand USD during a major emergency.

These amounts apply to either reservation strategy because the support depends on the scale of emergency services required, not the halls selected.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: One central hall	A_2: Two neighborhood halls	
States of nature	S_1: Mild emergency	S_2: Serious emergency	S_3: Major emergency
Uncertain event	Emergency level realized after the reservation decision		
Consequences	Fixed payoff in thousand USD tied only to the emergency state		



C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{emergency-state fixed payoff}$$

Alternative	Mild emergency	Serious emergency	Major emergency
One central hall	18	35	50
Two neighborhood halls	18	35	50

Alternative	Mild emergency	Serious emergency	Major emergency
One central hall	18	35	50
Two neighborhood halls	18	35	50



4 Two-Component Models

Problems:

4.1. Recycling Collection Contract (C1–C2)
4.1. Museum Ticket Platform (C1–C2)
4.2. Local Printing Contracts (C1–C2)
4.2. Cold-Storage Contracts (C1–C2)
4.3. Urban Landscaping Bids (C1–C2)
4.3. Solar Installation Bids (C1–C2)
4.4. Regional Bus Charters (C1–C2)
4.4. Air Cargo Charters (C1–C2)
4.5. Fruit Export Permits (C1–C2)
4.5. Seafood Export Licenses (C1–C2)
4.6. Warehouse Security Agreements (C1–C2)
4.6. Harbor Inspection Agreements (C1–C2)

4.1 Alternative-Based Variable Plus State-Based Variable

Problems:

4.1. Recycling Collection Contract (C1–C2)
4.1. Museum Ticket Platform (C1–C2)

Recycling Collection Contract (C1–C2)

A municipal contractor must choose an electric or diesel fleet for a two-month recycling collection contract. Supplier quotes and operating budgets indicate that each item collected by the electric fleet earns 0.6 thousand USD after fleet expenses, while each item collected by the diesel fleet earns 0.4 thousand USD. Revenue from recovered material adds 0.3 thousand USD for every item when recovery is high and 0.1 thousand USD when recovery is low.

The electric fleet is expected to collect 80 items under high recovery and 55 under low recovery. The diesel fleet is expected to collect 95 items under high recovery and 65 under low recovery. The diesel fleet processes more items, while the electric fleet earns more from each collection before recovery revenue is included.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Electric fleet	A_2: Diesel fleet
States of nature	S_1: High recovery	S_2: Low recovery
Uncertain event	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Electric fleet.
A_2	—	Alternative: Diesel fleet.
S_1	—	State: High recovery.
S_2	—	State: Low recovery.
$q(A_1, S_1)$	80 items	Quantity for Electric fleet under High recovery.
$q(A_1, S_2)$	55 items	Quantity for Electric fleet under Low recovery.
$q(A_2, S_1)$	95 items	Quantity for Diesel fleet under High recovery.
$q(A_2, S_2)$	65 items	Quantity for Diesel fleet under Low recovery.
$v(A_1)$	0.6 thousand USD per item	Fleet earnings after expenses for Electric fleet.
$v(A_2)$	0.4 thousand USD per item	Fleet earnings after expenses for Diesel fleet.
$v(S_1)$	0.3 thousand USD per item	Recovered-material revenue for High recovery.
$v(S_2)$	0.1 thousand USD per item	Recovered-material revenue for Low recovery.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big].$$

Recycling Collection Contract: payoff calculations.

All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times$ $\big[v(A_1)+v(S_1)\big]$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2)\times$ $\big[v(A_1)+v(S_2)\big]$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1)\times$ $\big[v(A_2)+v(S_1)\big]$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2)\times$ $\big[v(A_2)+v(S_2)\big]$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$80\times$ $\big[0.6+0.3\big]$ $=72$	$55\times$ $\big[0.6+0.1\big]$ $=38.5$
A_2	$95\times$ $\big[0.4+0.3\big]$ $=66.5$	$65\times$ $\big[0.4+0.1\big]$ $=32.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	72	38.5
A_2	66.5	32.5

Museum Ticket Platform (C1–C2)

A museum consortium must select a mobile, kiosk, or hybrid ticket platform. Its contracts measure activity in batches of one thousand admissions. After platform operating costs, each batch processed on the mobile platform earns 0.5 thousand USD, while each batch processed on either the kiosk or hybrid platform earns 0.4 thousand USD. Seasonal admission pricing adds 0.2 thousand USD to the earnings from every batch in peak season and 0.1 thousand USD in regular season. These figures come from recent admission volumes, vendor quotes, and processing-revenue forecasts.

The mobile platform is expected to process 70 batches during peak season and 45 during regular season. The kiosk platform is expected to process 60 and 50 batches, respectively. The hybrid platform is expected to process 65 batches during peak season and 55 during regular season. The mobile platform earns the most from each batch before seasonal admission revenue is included, while the hybrid platform is forecast to process the most batches during regular season.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Mobile platform	A_2: Kiosk platform	A_3: Hybrid platform
States of nature	S_1: Peak season	S_2: Regular season	
Uncertain event	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Mobile platform.
A_2	—	Alternative: Kiosk platform.
A_3	—	Alternative: Hybrid platform.
S_1	—	State: Peak season.
S_2	—	State: Regular season.
$q(A_1, S_1)$	70 batches	Quantity for Mobile platform under Peak season.
$q(A_1, S_2)$	45 batches	Quantity for Mobile platform under Regular season.
$q(A_2, S_1)$	60 batches	Quantity for Kiosk platform under Peak season.
$q(A_2, S_2)$	50 batches	Quantity for Kiosk platform under Regular season.
$q(A_3, S_1)$	65 batches	Quantity for Hybrid platform under Peak season.
$q(A_3, S_2)$	55 batches	Quantity for Hybrid platform under Regular season.
$v(A_1)$	0.5 thousand USD per batch of 1,000 admissions	Platform earnings after costs for Mobile platform.
$v(A_2)$	0.4 thousand USD per batch of 1,000 admissions	Platform earnings after costs for Kiosk platform.
$v(A_3)$	0.4 thousand USD per batch of 1,000 admissions	Platform earnings after costs for Hybrid platform.
$v(S_1)$	0.2 thousand USD per batch of 1,000 admissions	Seasonal admission earnings for Peak season.
$v(S_2)$	0.1 thousand USD per batch of 1,000 admissions	Seasonal admission earnings for Regular season.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big].$$

Museum Ticket Platform: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $=P_{ay}(A_2,S_2)$
A_3	$q(A_3,S_1)\times$ $[v(A_3)+v(S_1)]$ $=P_{ay}(A_3,S_1)$	$q(A_3,S_2)\times$ $[v(A_3)+v(S_2)]$ $=P_{ay}(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$70\times$ $[0.5+0.2]$ $=49$	$45\times$ $[0.5+0.1]$ $=27$
A_2	$60\times$ $[0.4+0.2]$ $=36$	$50\times$ $[0.4+0.1]$ $=25$
A_3	$65\times$ $[0.4+0.2]$ $=39$	$55\times$ $[0.4+0.1]$ $=27.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	49	27
A_2	36	25
A_3	39	27.5



4.2 Alternative-Based Variable Plus Alternative-Based Fixed

Problems:

4.2. Local Printing Contracts (C1–C2)

4.2. Cold-Storage Contracts (C1–C2)

Local Printing Contracts (C1–C2)

A local printing shop must choose a small or digital printer setup for school material packets. Contract terms provide 12 thousand USD of setup income for the small printer and 20 thousand USD for the digital printer. After ink, paper, and labor costs, the small printer earns 0.05 thousand USD per packet and the digital printer earns 0.04 thousand USD. Demand forecasts come from previous school orders.

The small printer is expected to produce 500 packets under high demand and 300 under low demand. The digital printer is expected to produce 700 and 400 packets, respectively.

The small setup earns more per packet, while the digital setup receives more setup income and can process more packets under both demand conditions.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Small printer	A_2: Digital printer
States of nature	S_1: High demand	S_2: Low demand
Uncertain event	Packet demand realized after the printing setup decision	
Consequences	Profit in thousand USD = packets × profit per packet + fixed profit	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Small printer.
A_2	—	Alternative: Digital printer.
S_1	—	State: High demand.
S_2	—	State: Low demand.
$q(A_1, S_1)$	500 packets	Quantity for Small printer under High demand.
$q(A_1, S_2)$	300 packets	Quantity for Small printer under Low demand.
$q(A_2, S_1)$	700 packets	Quantity for Digital printer under High demand.
$q(A_2, S_2)$	400 packets	Quantity for Digital printer under Low demand.
$v(A_1)$	0.05 thousand USD per packet	Printer earnings after costs for Small printer.
$v(A_2)$	0.04 thousand USD per packet	Printer earnings after costs for Digital printer.
$f(A_1)$	12 thousand USD	Setup income for Small printer.
$f(A_2)$	20 thousand USD	Setup income for Digital printer.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i).$$

Local Printing Contracts: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times v(A_1)$ $+f(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(A_1)$ $+f(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times v(A_2)$ $+f(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(A_2)$ $+f(A_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	500×0.05 $+12$ $=37$	300×0.05 $+12$ $=27$
A_2	700×0.04 $+20$ $=48$	400×0.04 $+20$ $=36$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	37	27
A_2	48	36



Cold-Storage Contracts (C1–C2)

A logistics company must choose a small, medium, or large cold-storage facility for agricultural pallets. Existing contracts provide 40 thousand USD of facility income for the small site, 25 thousand USD for the medium site, and 10 thousand USD for the large site. After operating costs, each stored pallet earns 0.18 thousand USD at the small facility, 0.12 thousand USD at the medium facility, and 0.08 thousand USD at the large facility. Volume forecasts use previous contracts and projected demand.

The small facility is expected to store 300 pallets under high demand and 220 under low demand. The medium facility is expected to store 600 and 420 pallets, respectively.

The large facility is expected to store 900 pallets under high demand and 700 under low demand. Larger facilities handle more pallets, while smaller facilities receive more contract income and earn more per pallet.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Small facility	A_2: Medium facility	A_3: Large facility
States of nature	S_1: High demand	S_2: Low demand	
Uncertain event	Storage demand realized after the lease decision		
Consequences	Profit in thousand USD = pallets × profit per pallet + fixed profit		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Small facility.
A_2	—	Alternative: Medium facility.
A_3	—	Alternative: Large facility.
S_1	—	State: High demand.
S_2	—	State: Low demand.
$q(A_1, S_1)$	300 pallets	Quantity for Small facility under High demand.
$q(A_1, S_2)$	220 pallets	Quantity for Small facility under Low demand.
$q(A_2, S_1)$	600 pallets	Quantity for Medium facility under High demand.
$q(A_2, S_2)$	420 pallets	Quantity for Medium facility under Low demand.
$q(A_3, S_1)$	900 pallets	Quantity for Large facility under High demand.
$q(A_3, S_2)$	700 pallets	Quantity for Large facility under Low demand.
$v(A_1)$	0.18 thousand USD per pallet	Storage earnings after costs for Small facility.
$v(A_2)$	0.12 thousand USD per pallet	Storage earnings after costs for Medium facility.
$v(A_3)$	0.08 thousand USD per pallet	Storage earnings after costs for Large facility.
$f(A_1)$	40 thousand USD	Facility-contract income for Small facility.
$f(A_2)$	25 thousand USD	Facility-contract income for Medium facility.
$f(A_3)$	10 thousand USD	Facility-contract income for Large facility.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i).$$

Cold-Storage Contracts: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times v(A_1)$ $+f(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(A_1)$ $+f(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times v(A_2)$ $+f(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(A_2)$ $+f(A_2)$ $=P_{ay}(A_2,S_2)$
A_3	$q(A_3,S_1)\times v(A_3)$ $+f(A_3)$ $=P_{ay}(A_3,S_1)$	$q(A_3,S_2)\times v(A_3)$ $+f(A_3)$ $=P_{ay}(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	300×0.18 $+40$ $=94$	220×0.18 $+40$ $=79.6$
A_2	600×0.12 $+25$ $=97$	420×0.12 $+25$ $=75.4$
A_3	900×0.08 $+10$ $=82$	700×0.08 $+10$ $=66$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	94	79.6
A_2	97	75.4
A_3	82	66



4.3 Alternative-Based Variable Plus State-Based Fixed

Problems:

4.3. Urban Landscaping Bids (C1–C2)
4.3. Solar Installation Bids (C1–C2)

Urban Landscaping Bids (C1–C2)

A landscaping company must focus its bids on residential or commercial city maintenance projects. Each completed project earns a team-specific profit, and grant funding or unreimbursed costs also affect the result. The estimates are based on previous contracts, projected labor and equipment costs, and expected project demand.

The residential team earns 0.6 thousand USD per project and is expected to complete 40 projects if the grant is available and 25 if no grant is available. The commercial team earns 1.0 thousand USD per project and is expected to complete 30 and 20 projects under the same respective states.

If the grant is available, it contributes 10 thousand USD toward either team's contract profit. Without the grant, either team must absorb 5 thousand USD of unreimbursed equipment and mobilization costs. The residential focus produces more projects, while the commercial focus earns more per project.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Residential	A_2: Commercial
States of nature	S_1: Grant available	S_2: No grant
Uncertain event	Grant outcome affecting the fixed profit adjustment	
Consequences	Profit in thousand USD = projects × profit per project + fixed profit	

C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Residential.
A_2	—	Alternative: Commercial.
S_1	—	State: Grant available.
S_2	—	State: No grant.
$q(A_1, S_1)$	40 projects	Quantity for Residential under Grant available.
$q(A_1, S_2)$	25 projects	Quantity for Residential under No grant.
$q(A_2, S_1)$	30 projects	Quantity for Commercial under Grant available.
$q(A_2, S_2)$	20 projects	Quantity for Commercial under No grant.
$v(A_1)$	0.6 thousand USD per project	Team profit for Residential.
$v(A_2)$	1.0 thousand USD per project	Team profit for Commercial.
$f(S_1)$	10 thousand USD	Grant funding or unreimbursed equipment and mobilization costs for Grant available.
$f(S_2)$	-5 thousand USD	Grant funding or unreimbursed equipment and mobilization costs for No grant. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(S_j).$$

Urban Landscaping Bids: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
	$q(A_1, S_1) \times v(A_1)$	$q(A_1, S_2) \times v(A_1)$
A_1	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$q(A_2, S_1) \times v(A_2)$	$q(A_2, S_2) \times v(A_2)$
A_2	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	40×0.6	25×0.6
A_1	$+10$	-5
	$= 34$	$= 10$
	30×1.0	20×1.0
A_2	$+10$	-5
	$= 40$	$= 15$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	34	10
A_2	40	15



Solar Installation Bids (C1–C2)

A solar company must choose a residential, commercial, or mixed portfolio. Each installation earns a portfolio-specific profit, and the government rebate either contributes funding or leaves costs unreimbursed. The estimates reflect recent bids, projected installation costs, and expected customer demand.

The residential focus earns 0.9 thousand USD per installation and is expected to complete 80 installations if the rebate continues and 50 if it ends. The commercial focus earns 1.6 thousand USD per installation, with projected volumes of 55 and 40 installations under the same respective states. The mixed portfolio earns 1.2 thousand USD per installation and is expected to complete 70 installations if the rebate continues and 45 if it ends.

If the rebate continues, it contributes 20 thousand USD to the profit of any portfolio. If it ends, any portfolio must absorb 10 thousand USD of administrative and equipment costs that would otherwise have been reimbursed. The residential strategy produces the largest volume, the commercial strategy earns the most per installation, and the mixed portfolio balances both features.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Residential	A_2: Commercial	A_3: Mixed portfolio
States of nature	S_1: Rebate continues	S_2: Rebate ends	
Uncertain event	Policy outcome affecting the fixed profit adjustment		
Consequences	Profit in thousand USD = installs × profit per install + fixed profit		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Residential.
A_2	—	Alternative: Commercial.
A_3	—	Alternative: Mixed portfolio.
S_1	—	State: Rebate continues.
S_2	—	State: Rebate ends.
$q(A_1, S_1)$	80 installations	Quantity for Residential under Rebate continues.
$q(A_1, S_2)$	50 installations	Quantity for Residential under Rebate ends.
$q(A_2, S_1)$	55 installations	Quantity for Commercial under Rebate continues.
$q(A_2, S_2)$	40 installations	Quantity for Commercial under Rebate ends.
$q(A_3, S_1)$	70 installations	Quantity for Mixed portfolio under Rebate continues.
$q(A_3, S_2)$	45 installations	Quantity for Mixed portfolio under Rebate ends.
$v(A_1)$	0.9 thousand USD per installation	Portfolio profit for Residential.
$v(A_2)$	1.6 thousand USD per installation	Portfolio profit for Commercial.
$v(A_3)$	1.2 thousand USD per installation	Portfolio profit for Mixed portfolio.
$f(S_1)$	20 thousand USD	Rebate funding or unreimbursed administrative and equipment costs for Rebate continues.
$f(S_2)$	-10 thousand USD	Rebate funding or unreimbursed administrative and equipment costs for Rebate ends. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(S_j).$$

Solar Installation Bids: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
	$q(A_1, S_1) \times v(A_1)$	$q(A_1, S_2) \times v(A_1)$
A_1	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$q(A_2, S_1) \times v(A_2)$	$q(A_2, S_2) \times v(A_2)$
A_2	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$
	$q(A_3, S_1) \times v(A_3)$	$q(A_3, S_2) \times v(A_3)$
A_3	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_3, S_1)$	$= P_{ay}(A_3, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	80×0.9	50×0.9
A_1	$+ 20$	$- 10$
	$= 92$	$= 35$
	55×1.6	40×1.6
A_2	$+ 20$	$- 10$
	$= 108$	$= 54$
	70×1.2	45×1.2
A_3	$+ 20$	$- 10$
	$= 104$	$= 44$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	92	35
A_2	108	54
A_3	104	44



4.4 State-Based Variable Plus Alternative-Based Fixed

Problems:

4.4. Regional Bus Charters (C1–C2)
4.4. Air Cargo Charters (C1–C2)

Regional Bus Charters (C1–C2)

A regional transport firm must choose a mini-bus or coach fleet for school-trip charters. Fleet contracts provide 15 thousand USD of income for mini-buses and 25 thousand USD for coaches, in addition to the profit from completed charters. The estimates are based on previous school contracts, expected bookings, and projected fuel and operating costs.

The mini-bus fleet is expected to complete 25 charters under regular fuel costs and 20 under high fuel costs. The coach fleet has projected volumes of 18 and 16 charters under the same respective states.

Profit per charter is 4 thousand USD when fuel costs are regular and 2 thousand USD when fuel costs are high. The mini-bus fleet completes more charters, while the coach fleet receives more contract income.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Mini-bus fleet	A_2: Coach fleet
States of nature	S_1: Regular fuel	S_2: High fuel
Uncertain event	Fuel costs affecting profit per charter	
Consequences	Profit in thousand USD = charters × profit per charter + fixed profit	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Mini-bus fleet.
A_2	—	Alternative: Coach fleet.
S_1	—	State: Regular fuel.
S_2	—	State: High fuel.
$q(A_1, S_1)$	25 charters	Quantity for Mini-bus fleet under Regular fuel.
$q(A_1, S_2)$	20 charters	Quantity for Mini-bus fleet under High fuel.
$q(A_2, S_1)$	18 charters	Quantity for Coach fleet under Regular fuel.
$q(A_2, S_2)$	16 charters	Quantity for Coach fleet under High fuel.
$v(S_1)$	4 thousand USD per charter	Fuel-dependent charter profit for Regular fuel.
$v(S_2)$	2 thousand USD per charter	Fuel-dependent charter profit for High fuel.
$f(A_1)$	15 thousand USD	Fleet-contract income for Mini-bus fleet.
$f(A_2)$	25 thousand USD	Fleet-contract income for Coach fleet.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i).$$

Regional Bus Charters: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
	$q(A_1, S_1) \times v(S_1)$	$q(A_1, S_2) \times v(S_2)$
A_1	$+ f(A_1)$	$+ f(A_1)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$q(A_2, S_1) \times v(S_1)$	$q(A_2, S_2) \times v(S_2)$
A_2	$+ f(A_2)$	$+ f(A_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	25×4	20×2
A_1	$+15$	$+15$
	$= 115$	$= 55$
	18×4	16×2
A_2	$+25$	$+25$
	$= 97$	$= 57$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	115	55
A_2	97	57



Air Cargo Charters (C1–C2)

An air cargo firm must lease narrow-body or wide-body aircraft. Lease contracts provide 30 thousand USD of income for narrow-body aircraft and 50 thousand USD for wide-body aircraft, in addition to profit from completed charters. The estimates reflect previous contracts, expected freight demand, lease costs, and projected fuel expenses.

Narrow-body aircraft are expected to complete 20 charters under low fuel costs, 18 under medium fuel costs, and 15 under high fuel costs. Wide-body aircraft have projected volumes of 14, 12, and 9 charters under the same respective states.

Profit per charter is 8 thousand USD under low fuel costs, 5 thousand USD under medium fuel costs, and 2 thousand USD under high fuel costs. Narrow-body aircraft complete more charters, while wide-body aircraft receive more lease-contract income.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Narrow-body lease	A_2: Wide-body lease	
States of nature	S_1: Low fuel	S_2: Medium fuel	S_3: High fuel
Uncertain event	Fuel prices affecting profit per charter		
Consequences	Profit in thousand USD = charters × profit per charter + fixed profit		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Narrow-body lease.
A_2	—	Alternative: Wide-body lease.
S_1	—	State: Low fuel.
S_2	—	State: Medium fuel.
S_3	—	State: High fuel.
$q(A_1, S_1)$	20 charters	Quantity for Narrow-body lease under Low fuel.
$q(A_1, S_2)$	18 charters	Quantity for Narrow-body lease under Medium fuel.
$q(A_1, S_3)$	15 charters	Quantity for Narrow-body lease under High fuel.
$q(A_2, S_1)$	14 charters	Quantity for Wide-body lease under Low fuel.
$q(A_2, S_2)$	12 charters	Quantity for Wide-body lease under Medium fuel.
$q(A_2, S_3)$	9 charters	Quantity for Wide-body lease under High fuel.
$v(S_1)$	8 thousand USD per charter	Fuel-dependent charter profit for Low fuel.
$v(S_2)$	5 thousand USD per charter	Fuel-dependent charter profit for Medium fuel.
$v(S_3)$	2 thousand USD per charter	Fuel-dependent charter profit for High fuel.
$f(A_1)$	30 thousand USD	Lease-contract income for Narrow-body lease.
$f(A_2)$	50 thousand USD	Lease-contract income for Wide-body lease.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i).$$

Air Cargo Charters: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
	$q(A_1,S_1)\times v(S_1)$	$q(A_1,S_2)\times v(S_2)$	$q(A_1,S_3)\times v(S_3)$
A_1	$+f(A_1)$	$+f(A_1)$	$+f(A_1)$
	$=P_{ay}(A_1,S_1)$	$=P_{ay}(A_1,S_2)$	$=P_{ay}(A_1,S_3)$
	$q(A_2,S_1)\times v(S_1)$	$q(A_2,S_2)\times v(S_2)$	$q(A_2,S_3)\times v(S_3)$
A_2	$+f(A_2)$	$+f(A_2)$	$+f(A_2)$
	$=P_{ay}(A_2,S_1)$	$=P_{ay}(A_2,S_2)$	$=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
	20×8	18×5	15×2
A_1	$+30$	$+30$	$+30$
	$=190$	$=120$	$=60$
	14×8	12×5	9×2
A_2	$+50$	$+50$	$+50$
	$=162$	$=110$	$=68$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	190	120	60
A_2	162	110	68

4.5 State-Based Variable Plus State-Based Fixed

Problems:

4.5. Fruit Export Permits (C1–C2)
4.5. Seafood Export Licenses (C1–C2)

Fruit Export Permits (C1–C2)

A fruit exporter must choose whether to focus its permit on citrus or berry exports. Profit comes from each crate exported, together with supply-contract income or emergency sourcing costs determined by the harvest. The estimates are based on expected export volumes, seasonal yields, transportation costs, and market demand.

During a strong harvest, favorable supply contracts contribute 12 thousand USD to the exporter's profit, and each crate earns 0.4 thousand USD. During a weak harvest, emergency sourcing costs reduce profit by 8 thousand USD, and each crate earns 0.2 thousand USD.

The citrus focus is expected to export 200 crates in a strong season and 150 in a weak season. The berry focus is expected to export 180 and 120 crates under the same respective conditions. Citrus maintains the larger modeled export volume in both states.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Citrus focus	A_2: Berry focus
States of nature	S_1: Strong	S_2: Weak
Uncertain event	Seasonal conditions affecting prices and fixed profit adjustments	
Consequences	Profit in thousand USD = crates × profit per crate + fixed profit	

C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Citrus focus.
A_2	—	Alternative: Berry focus.
S_1	—	State: Strong.
S_2	—	State: Weak.
$q(A_1, S_1)$	200 crates	Quantity for Citrus focus under Strong.
$q(A_1, S_2)$	150 crates	Quantity for Citrus focus under Weak.
$q(A_2, S_1)$	180 crates	Quantity for Berry focus under Strong.
$q(A_2, S_2)$	120 crates	Quantity for Berry focus under Weak.
$v(S_1)$	0.4 thousand USD per crate	Harvest-dependent export profit for Strong.
$v(S_2)$	0.2 thousand USD per crate	Harvest-dependent export profit for Weak.
$f(S_1)$	12 thousand USD	Supply-contract income or emergency sourcing costs for Strong.
$f(S_2)$	-8 thousand USD	Supply-contract income or emergency sourcing costs for Weak. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(S_j).$$

Fruit Export Permits: payoff calculations.

All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
	$q(A_1, S_1) \times v(S_1)$	$q(A_1, S_2) \times v(S_2)$
A_1	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$q(A_2, S_1) \times v(S_1)$	$q(A_2, S_2) \times v(S_2)$
A_2	$+ f(S_1)$	$+ f(S_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	200×0.4	150×0.2
A_1	$+12$	-8
	$= 92$	$= 22$
	180×0.4	120×0.2
A_2	$+12$	-8
	$= 84$	$= 16$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	92	22
A_2	84	16



Seafood Export Licenses (C1–C2)

A seafood exporter must choose between a frozen-fillet focus and a fresh-export focus. Profit comes from each crate exported, together with advance-purchase income or weather-related costs determined by the season. The estimates reflect expected demand, storage and transportation costs, previous export volumes, and possible weather disruptions.

During a strong tourism season, advance-purchase contracts contribute 30 thousand USD to profit, and each crate earns 0.7 thousand USD. During a normal season, those contracts contribute 10 thousand USD, and each crate earns 0.5 thousand USD. During a stormy season, weather-related storage and rescheduling costs reduce profit by 15 thousand USD, and each crate earns 0.2 thousand USD.

The frozen-fillet focus is expected to export 120 crates in a strong season, 100 in a normal season, and 80 in a stormy season. The fresh-export focus is expected to export 90, 110, and 70 crates under the same respective states. Frozen products perform better in strong and stormy conditions, while fresh exports have the larger modeled volume during a normal season.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Frozen fillet focus	A_2: Fresh export focus	
States of nature	S_1: Strong	S_2: Normal	S_3: Stormy
Uncertain event	Seasonal conditions affecting prices and fixed profit adjustments		
Consequences	Profit in thousand USD = crates × profit per crate + fixed profit		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Frozen fillet focus.
A_2	—	Alternative: Fresh export focus.
S_1	—	State: Strong.
S_2	—	State: Normal.
S_3	—	State: Stormy.
$q(A_1, S_1)$	120 crates	Quantity for Frozen fillet focus under Strong.
$q(A_1, S_2)$	100 crates	Quantity for Frozen fillet focus under Normal.
$q(A_1, S_3)$	80 crates	Quantity for Frozen fillet focus under Stormy.
$q(A_2, S_1)$	90 crates	Quantity for Fresh export focus under Strong.
$q(A_2, S_2)$	110 crates	Quantity for Fresh export focus under Normal.
$q(A_2, S_3)$	70 crates	Quantity for Fresh export focus under Stormy.
$v(S_1)$	0.7 thousand USD per crate	Seasonal export profit for Strong.
$v(S_2)$	0.5 thousand USD per crate	Seasonal export profit for Normal.
$v(S_3)$	0.2 thousand USD per crate	Seasonal export profit for Stormy.
$f(S_1)$	30 thousand USD	Advance-purchase income or weather-related storage and rescheduling costs for Strong.
$f(S_2)$	10 thousand USD	Advance-purchase income or weather-related storage and rescheduling costs for Normal.
$f(S_3)$	-15 thousand USD	Advance-purchase income or weather-related storage and rescheduling costs for Stormy. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(S_j).$$

Seafood Export Licenses: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
	$q(A_1,S_1)\times v(S_1)$	$q(A_1,S_2)\times v(S_2)$	$q(A_1,S_3)\times v(S_3)$
A_1	$+f(S_1)$	$+f(S_2)$	$+f(S_3)$
	$=P_{ay}(A_1,S_1)$	$=P_{ay}(A_1,S_2)$	$=P_{ay}(A_1,S_3)$
	$q(A_2,S_1)\times v(S_1)$	$q(A_2,S_2)\times v(S_2)$	$q(A_2,S_3)\times v(S_3)$
A_2	$+f(S_1)$	$+f(S_2)$	$+f(S_3)$
	$=P_{ay}(A_2,S_1)$	$=P_{ay}(A_2,S_2)$	$=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
	120×0.7	100×0.5	80×0.2
A_1	$+30$	$+10$	-15
	$=114$	$=60$	$=1$
	90×0.7	110×0.5	70×0.2
A_2	$+30$	$+10$	-15
	$=93$	$=65$	$=-1$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	114	60	1
A_2	93	65	-1

4.6 Alternative-Based Fixed Plus State-Based Fixed

Problems:

4.6. Warehouse Security Agreements (C1–C2)
4.6. Harbor Inspection Agreements (C1–C2)

Warehouse Security Agreements (C1–C2)

A warehouse operator must choose between a standard and an enhanced security agreement for protecting seasonal inventory. Contract terms provide 20 thousand USD of net income under the standard agreement and 35 thousand USD under the enhanced agreement. Inventory forecasts show that the normal season brings a further 6 thousand USD of security funding, while the festival season brings 14 thousand USD because larger inventories require greater security support.

These agreement amounts and seasonal funding are both included in the operator's financial result, whichever agreement is selected.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Standard security	A_2: Enhanced security
States of nature	S_1: Normal season	S_2: Festival season
Uncertain event	Demand-season state realized during the security period	
Consequences	Profit in thousand USD = agreement fixed payoff + state fixed payoff	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Standard security.
A_2	—	Alternative: Enhanced security.
S_1	—	State: Normal season.
S_2	—	State: Festival season.
$f(A_1)$	20 thousand USD	Security-agreement net income for Standard security.
$f(A_2)$	35 thousand USD	Security-agreement net income for Enhanced security.
$f(S_1)$	6 thousand USD	Seasonal security funding for Normal season.
$f(S_2)$	14 thousand USD	Seasonal security funding for Festival season.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = f(A_i) + f(S_j).$$

Warehouse Security Agreements: payoff calculations.

All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$20+6$ $=26$	$20+14$ $=34$
A_2	$35+6$ $=41$	$35+14$ $=49$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	26	34
A_2	41	49



Harbor Inspection Agreements (C1–C2)

A port authority must choose between a standard inspection agreement and a rapid-response inspection agreement. Contract terms provide 28 thousand USD of net income under the standard agreement and 42 thousand USD under the rapid-response agreement. Operations forecasts indicate that a calm season adds 12 thousand USD and a busy season adds 5 thousand USD, while storm repairs reduce the result by 10 thousand USD. These amounts apply whichever agreement is chosen.

The harbor estimates come from the inspection contracts, maintenance plans, and staffing requirements for each

operating condition.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Standard inspection	A_2: Rapid-response inspection	
States of nature	S_1: Calm season	S_2: Busy season	S_3: Storm-repair season
Uncertain event	Harbor condition state realized during the inspection period		
Consequences	Profit in thousand USD = agreement fixed payoff + state fixed payoff		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Standard inspection.
A_2	—	Alternative: Rapid-response inspection.
S_1	—	State: Calm season.
S_2	—	State: Busy season.
S_3	—	State: Storm-repair season.
$f(A_1)$	28 thousand USD	Inspection-agreement net income for Standard inspection.
$f(A_2)$	42 thousand USD	Inspection-agreement net income for Rapid-response inspection.
$f(S_1)$	12 thousand USD	Seasonal income or storm-repair costs for Calm season.
$f(S_2)$	5 thousand USD	Seasonal income or storm-repair costs for Busy season.
$f(S_3)$	-10 thousand USD	Seasonal income or storm-repair costs for Storm-repair season. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = f(A_i) + f(S_j).$$

Harbor Inspection Agreements: payoff calculations.

All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$	$f(A_1)+f(S_3)$ $=P_{ay}(A_1,S_3)$
A_2	$f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$	$f(A_2)+f(S_3)$ $=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$28+12$ $=40$	$28+5$ $=33$	$28-10$ $=18$
A_2	$42+12$ $=54$	$42+5$ $=47$	$42-10$ $=32$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	40	33	18
A_2	54	47	32



5 Three-Component Models

Problems:

5.1. Water Treatment Modules (C1–C2)

5.1. Freight Locker Network (C1–C2)

5.2. District Heating Supply (C1–C2)

5.2. Port Cold-Chain Service (C1–C2)

5.3. Pharmacy Delivery Plan (C1–C2)

5.3. Timber Processing Line (C1–C2)

5.4. Public Library Outreach (C1–C2)

5.4. Grain Terminal Schedule (C1–C2)

5.1 Alternative-Based Variable Plus State-Based Variable Plus Alternative-Based Fixed

Problems:

5.1. Water Treatment Modules (C1–C2)

5.1. Freight Locker Network (C1–C2)

Water Treatment Modules (C1–C2)

A utility must choose between a membrane unit and a filtration unit for treating batches of water delivered to remote communities. Supplier terms give the membrane unit 8 thousand USD of contract income plus earnings of 0.8 thousand USD for every batch treated. The filtration unit receives 12 thousand USD of contract income plus 0.6 thousand USD for every batch. Seasonal water tariffs add 0.3 thousand USD to the earnings from each batch in the dry season and 0.1 thousand USD in the wet season. The estimates come from supplier quotes, operating budgets, and demand forecasts.

Capacity studies predict that the membrane unit will treat 50 batches in the dry season and 35 in the wet season. They predict 60 and 45 batches, respectively, for the filtration unit.

The membrane unit earns more for each batch before seasonal tariffs, whereas the filtration unit receives more contract income and has greater forecast capacity.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Membrane unit	A_2: Filtration unit
States of nature	S_1: Dry season	S_2: Wet season
Uncertain event	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Membrane unit.
A_2	—	Alternative: Filtration unit.
S_1	—	State: Dry season.
S_2	—	State: Wet season.
$q(A_1, S_1)$	50 water batches	Quantity for Membrane unit under Dry season.
$q(A_1, S_2)$	35 water batches	Quantity for Membrane unit under Wet season.
$q(A_2, S_1)$	60 water batches	Quantity for Filtration unit under Dry season.
$q(A_2, S_2)$	45 water batches	Quantity for Filtration unit under Wet season.
$v(A_1)$	0.8 thousand USD per water batch	Treatment earnings for Membrane unit.
$v(A_2)$	0.6 thousand USD per water batch	Treatment earnings for Filtration unit.
$v(S_1)$	0.3 thousand USD per water batch	Seasonal water-tariff earnings for Dry season.
$v(S_2)$	0.1 thousand USD per water batch	Seasonal water-tariff earnings for Wet season.
$f(A_1)$	8 thousand USD	Supplier-contract income for Membrane unit.
$f(A_2)$	12 thousand USD	Supplier-contract income for Filtration unit.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i).$$

Water Treatment Modules: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $+f(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $+f(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $+f(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $+f(A_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$50\times$ $[0.8+0.3]$ $+8$ $=63$	$35\times$ $[0.8+0.1]$ $+8$ $=39.5$
A_2	$60\times$ $[0.6+0.3]$ $+12$ $=66$	$45\times$ $[0.6+0.1]$ $+12$ $=43.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	63	39.5
A_2	66	43.5



Freight Locker Network (C1–C2)

A parcel operator must choose a metro-locker system, a suburban-locker system, or a mixed network. Lease proposals provide 10 thousand USD of contract income for the metro system, 14 thousand USD for the suburban system, and 12 thousand USD for the mixed network. After operating costs, metro and mixed lockers each earn 0.5 thousand USD per shipment batch, while suburban lockers earn 0.4 thousand USD. Strong demand adds 0.2 thousand USD to the earnings from every batch, and steady demand adds 0.1 thousand USD. The estimates come from lease terms, recent parcel volumes, and operating budgets.

Volume forecasts show metro lockers handling 80 batches under strong demand and 55 under steady demand, while suburban lockers handle 65 and 60 batches, respectively.

The mixed network is expected to handle 75 batches under strong demand and 65 under steady demand. The metro network has the highest modeled volume under strong demand, while the mixed network handles the most batches under steady demand.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Metro lockers	A_2: Suburban lockers	A_3: Mixed network
States of nature	S_1: Strong demand	S_2: Steady demand	
Uncertain event	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Metro lockers.
A_2	—	Alternative: Suburban lockers.
A_3	—	Alternative: Mixed network.
S_1	—	State: Strong demand.
S_2	—	State: Steady demand.
$q(A_1, S_1)$	80 shipment batches	Quantity for Metro lockers under Strong demand.
$q(A_1, S_2)$	55 shipment batches	Quantity for Metro lockers under Steady demand.
$q(A_2, S_1)$	65 shipment batches	Quantity for Suburban lockers under Strong demand.
$q(A_2, S_2)$	60 shipment batches	Quantity for Suburban lockers under Steady demand.
$q(A_3, S_1)$	75 shipment batches	Quantity for Mixed network under Strong demand.
$q(A_3, S_2)$	65 shipment batches	Quantity for Mixed network under Steady demand.
$v(A_1)$	0.5 thousand USD per shipment batch	Locker earnings after costs for Metro lockers.
$v(A_2)$	0.4 thousand USD per shipment batch	Locker earnings after costs for Suburban lockers.
$v(A_3)$	0.5 thousand USD per shipment batch	Locker earnings after costs for Mixed network.
$v(S_1)$	0.2 thousand USD per shipment batch	Demand-related earnings for Strong demand.
$v(S_2)$	0.1 thousand USD per shipment batch	Demand-related earnings for Steady demand.
$f(A_1)$	10 thousand USD	Lease-contract income for Metro lockers.
$f(A_2)$	14 thousand USD	Lease-contract income for Suburban lockers.
$f(A_3)$	12 thousand USD	Lease-contract income for Mixed network.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i).$$

Freight Locker Network: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $+f(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $+f(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $+f(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $+f(A_2)$ $=P_{ay}(A_2,S_2)$
A_3	$q(A_3,S_1)\times$ $[v(A_3)+v(S_1)]$ $+f(A_3)$ $=P_{ay}(A_3,S_1)$	$q(A_3,S_2)\times$ $[v(A_3)+v(S_2)]$ $+f(A_3)$ $=P_{ay}(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$80\times$ $[0.5+0.2]$ $+10$ $=66$	$55\times$ $[0.5+0.1]$ $+10$ $=43$
A_2	$65\times$ $[0.4+0.2]$ $+14$ $=53$	$60\times$ $[0.4+0.1]$ $+14$ $=44$
A_3	$75\times$ $[0.5+0.2]$ $+12$ $=64.5$	$65\times$ $[0.5+0.1]$ $+12$ $=51$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	66	43
A_2	53	44
A_3	64.5	51



5.2 Alternative-Based Variable Plus State-Based Variable Plus State-Based Fixed

Problems:

5.2. District Heating Supply (C1–C2)

5.2. Port Cold-Chain Service (C1–C2)

District Heating Supply (C1–C2)

A district energy provider must choose between biomass heat and heat pumps. Energy-price contracts and operating budgets show earnings of 0.5 thousand USD for each biomass heating unit and 0.6 thousand USD for each heat-pump unit. Winter tariffs add 0.1 thousand USD for every unit in a mild winter and 0.3 thousand USD in a cold winter. A mild-winter service subsidy contributes another 6 thousand USD, while cold-weather emergency expenses reduce profit by 4 thousand USD. Demand forecasts supply the expected volumes.

Biomass heat contributes 0.5 thousand USD per unit and is expected to supply 55 units during a mild winter and 85 during a cold winter. Heat pumps contribute 0.6 thousand USD per unit, with projected volumes of 65 and 95 units under the same respective states.

Heat pumps provide the higher payoff per unit and the larger modeled supply in both winter conditions.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Biomass heat	A_2: Heat pumps
States of nature	S_1: Mild winter	S_2: Cold winter
Uncertain event	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Biomass heat.
A_2	—	Alternative: Heat pumps.
S_1	—	State: Mild winter.
S_2	—	State: Cold winter.
$q(A_1, S_1)$	55 heating units	Quantity for Biomass heat under Mild winter.
$q(A_1, S_2)$	85 heating units	Quantity for Biomass heat under Cold winter.
$q(A_2, S_1)$	65 heating units	Quantity for Heat pumps under Mild winter.
$q(A_2, S_2)$	95 heating units	Quantity for Heat pumps under Cold winter.
$v(A_1)$	0.5 thousand USD per heating unit	Heating earnings for Biomass heat.
$v(A_2)$	0.6 thousand USD per heating unit	Heating earnings for Heat pumps.
$v(S_1)$	0.1 thousand USD per heating unit	Winter-tariff earnings for Mild winter.
$v(S_2)$	0.3 thousand USD per heating unit	Winter-tariff earnings for Cold winter.
$f(S_1)$	6 thousand USD	Service subsidy or emergency expenses for Mild winter.
$f(S_2)$	-4 thousand USD	Service subsidy or emergency expenses for Cold winter. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(S_j).$$

District Heating Supply: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $+f(S_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$55\times$ $[0.5+0.1]$ $+6$ $=39$	$85\times$ $[0.5+0.3]$ -4 $=64$
A_2	$65\times$ $[0.6+0.1]$ $+6$ $=51.5$	$95\times$ $[0.6+0.3]$ -4 $=81.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	39	64
A_2	51.5	81.5



Port Cold-Chain Service (C1–C2)

A port authority must choose between automated and crew-operated cold-chain service. Cost studies show that automation earns 0.7 thousand USD per refrigerated container batch and crew service earns 0.5 thousand USD. Clear flow adds 0.2 thousand USD to the earnings from each batch and provides 8 thousand USD of scheduling savings. Congestion adds 0.1 thousand USD per batch but creates 3 thousand USD of extra coordination costs. An inspection surge adds 0.3 thousand USD per batch and provides 5 thousand USD of inspection-service income. Volume estimates come from earlier congestion and inspection records.

The automated service contributes 0.7 thousand USD per batch and is expected to handle 80 batches under clear flow, 60 under congestion, and 45 during an inspection surge. The crew service contributes 0.5 thousand USD per batch, with projected volumes of 70, 65, and 55 batches under the same respective states.

Automation handles more batches under clear flow, while the crew service maintains higher volumes under congestion and inspection surges.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Automated service	A_2: Crew service	
States of nature	S_1: Clear flow	S_2: Congestion	S_3: Inspection surge
Uncertain event	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Automated service.
A_2	—	Alternative: Crew service.
S_1	—	State: Clear flow.
S_2	—	State: Congestion.
S_3	—	State: Inspection surge.
$q(A_1, S_1)$	80 container batches	Quantity for Automated service under Clear flow.
$q(A_1, S_2)$	60 container batches	Quantity for Automated service under Congestion.
$q(A_1, S_3)$	45 container batches	Quantity for Automated service under Inspection surge.
$q(A_2, S_1)$	70 container batches	Quantity for Crew service under Clear flow.
$q(A_2, S_2)$	65 container batches	Quantity for Crew service under Congestion.
$q(A_2, S_3)$	55 container batches	Quantity for Crew service under Inspection surge.
$v(A_1)$	0.7 thousand USD per container batch	Service earnings for Automated service.
$v(A_2)$	0.5 thousand USD per container batch	Service earnings for Crew service.
$v(S_1)$	0.2 thousand USD per container batch	Flow-related earnings for Clear flow.
$v(S_2)$	0.1 thousand USD per container batch	Flow-related earnings for Congestion.
$v(S_3)$	0.3 thousand USD per container batch	Flow-related earnings for Inspection surge.
$f(S_1)$	8 thousand USD	Scheduling savings, coordination costs, or inspection-service income for Clear flow.
$f(S_2)$	-3 thousand USD	Scheduling savings, coordination costs, or inspection-service income for Congestion. Negative: reduces the payoff.
$f(S_3)$	5 thousand USD	Scheduling savings, coordination costs, or inspection-service income for Inspection surge.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(S_j).$$

Port Cold-Chain Service: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $+f(S_2)$ $=P_{ay}(A_1,S_2)$	$q(A_1,S_3)\times$ $[v(A_1)+v(S_3)]$ $+f(S_3)$ $=P_{ay}(A_1,S_3)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $+f(S_2)$ $=P_{ay}(A_2,S_2)$	$q(A_2,S_3)\times$ $[v(A_2)+v(S_3)]$ $+f(S_3)$ $=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$80\times$ $[0.7+0.2]$ $+8$ $=80$	$60\times$ $[0.7+0.1]$ -3 $=45$	$45\times$ $[0.7+0.3]$ $+5$ $=50$
A_2	$70\times$ $[0.5+0.2]$ $+8$ $=57$	$65\times$ $[0.5+0.1]$ -3 $=36$	$55\times$ $[0.5+0.3]$ $+5$ $=49$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	80	45	50
A_2	57	36	49

5.3 Alternative-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

Problems:

5.3. Pharmacy Delivery Plan (C1–C2)
5.3. Timber Processing Line (C1–C2)

Pharmacy Delivery Plan (C1–C2)

A pharmacy cooperative must choose cargo bikes or electric vans for prescription deliveries. The bike contract provides 9 thousand USD of route income and earns 0.6 thousand USD for each batch delivered. The van contract provides 4 thousand USD of route income and earns 0.8 thousand USD per batch. Recent traffic records also indicate 6 thousand USD of scheduling savings in light traffic, while delays in heavy traffic impose 5 thousand USD of extra costs on either plan.

Capacity forecasts estimate that cargo bikes will deliver 60 batches in light traffic and 45 in heavy traffic. Electric vans are expected to deliver 75 and 55 batches, respectively.

The estimates come from vehicle costs, route contracts, delivery capacity, and recent traffic patterns. Vans earn more per batch and carry more batches, whereas bikes receive more route income.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Cargo bikes	A_2: Electric vans
States of nature	S_1: Light traffic	S_2: Heavy traffic
Uncertain event	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Cargo bikes.
A_2	—	Alternative: Electric vans.
S_1	—	State: Light traffic.
S_2	—	State: Heavy traffic.
$q(A_1, S_1)$	60 delivery batches	Quantity for Cargo bikes under Light traffic.
$q(A_1, S_2)$	45 delivery batches	Quantity for Cargo bikes under Heavy traffic.
$q(A_2, S_1)$	75 delivery batches	Quantity for Electric vans under Light traffic.
$q(A_2, S_2)$	55 delivery batches	Quantity for Electric vans under Heavy traffic.
$v(A_1)$	0.6 thousand USD per delivery batch	Delivery earnings for Cargo bikes.
$v(A_2)$	0.8 thousand USD per delivery batch	Delivery earnings for Electric vans.
$f(A_1)$	9 thousand USD	Route-contract income for Cargo bikes.
$f(A_2)$	4 thousand USD	Route-contract income for Electric vans.
$f(S_1)$	6 thousand USD	Scheduling savings or delay costs for Light traffic.
$f(S_2)$	-5 thousand USD	Scheduling savings or delay costs for Heavy traffic. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i) + f(S_j).$$

Pharmacy Delivery Plan: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times v(A_1)$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(A_1)$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times v(A_2)$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(A_2)$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	60×0.6 $+9+6$ $=51$	45×0.6 $+9-5$ $=31$
A_2	75×0.8 $+4+6$ $=70$	55×0.8 $+4-5$ $=43$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	51	31
A_2	70	43



Timber Processing Line (C1–C2)

A wood cooperative must choose a precision or standard lumber-processing line. The precision line's sales agreement provides 7 thousand USD of income and earnings of 0.9 thousand USD per batch. The standard line's agreement provides 13 thousand USD and earnings of 0.6 thousand USD per batch. Market contracts add 10 thousand USD when the market is firm and 2 thousand USD when it is normal, while weak-market storage costs reduce profit by 6 thousand USD.

Production records and demand forecasts indicate that the precision line will process 70 batches in a firm market, 55 in a normal market, and 40 in a weak market. The standard line is expected to process 85, 65, and 50 batches under those respective conditions.

The precision line earns more per batch, while the standard line receives more contract income and processes more batches in every market condition.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Precision line	A_2: Standard line	
States of nature	S_1: Firm market	S_2: Normal market	S_3: Weak market
Uncertain event	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Precision line.
A_2	—	Alternative: Standard line.
S_1	—	State: Firm market.
S_2	—	State: Normal market.
S_3	—	State: Weak market.
$q(A_1, S_1)$	70 lumber batches	Quantity for Precision line under Firm market.
$q(A_1, S_2)$	55 lumber batches	Quantity for Precision line under Normal market.
$q(A_1, S_3)$	40 lumber batches	Quantity for Precision line under Weak market.
$q(A_2, S_1)$	85 lumber batches	Quantity for Standard line under Firm market.
$q(A_2, S_2)$	65 lumber batches	Quantity for Standard line under Normal market.
$q(A_2, S_3)$	50 lumber batches	Quantity for Standard line under Weak market.
$v(A_1)$	0.9 thousand USD per lumber batch	Processing earnings for Precision line.
$v(A_2)$	0.6 thousand USD per lumber batch	Processing earnings for Standard line.
$f(A_1)$	7 thousand USD	Sales-agreement income for Precision line.
$f(A_2)$	13 thousand USD	Sales-agreement income for Standard line.
$f(S_1)$	10 thousand USD	Market-contract income or storage costs for Firm market.
$f(S_2)$	2 thousand USD	Market-contract income or storage costs for Normal market.
$f(S_3)$	-6 thousand USD	Market-contract income or storage costs for Weak market. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i) + f(S_j).$$

Timber Processing Line: payoff calculations.

All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$q(A_1,S_1)\times v(A_1)$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(A_1)$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$	$q(A_1,S_3)\times v(A_1)$ $+f(A_1)+f(S_3)$ $=P_{ay}(A_1,S_3)$
A_2	$q(A_2,S_1)\times v(A_2)$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(A_2)$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$	$q(A_2,S_3)\times v(A_2)$ $+f(A_2)+f(S_3)$ $=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	70×0.9 $+7+10$ $=80$	55×0.9 $+7+2$ $=58.5$	40×0.9 $+7-6$ $=37$
A_2	85×0.6 $+13+10$ $=74$	65×0.6 $+13+2$ $=54$	50×0.6 $+13-6$ $=37$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	80	58.5	37
A_2	74	54	37



5.4 State-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

Problems:

5.4. Public Library Outreach (C1–C2)
5.4. Grain Terminal Schedule (C1–C2)

Public Library Outreach (C1–C2)

A library system must choose a mobile library or pop-up branches. The program budget provides 8 thousand USD for the mobile library and 11 thousand USD for pop-up branches. When attendance is high, funding provides 0.7 thousand USD for each group served and a further 5 thousand USD for the program. When attendance is low, it provides 0.4 thousand USD per group, while unrecovered venue and publicity costs reduce the result by 2 thousand USD. The estimates come from earlier participation, operating costs, and the funding schedule.

Attendance forecasts show the mobile library serving 50 groups under high attendance and 30 under low attendance. Pop-up branches are expected to serve 45 and 35 groups, respectively.

The mobile library serves more groups under high attendance, while pop-up branches serve more under low attendance and receive the larger initial program allocation.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Mobile library	A_2: Pop-up branches
States of nature	S_1: High attendance	S_2: Low attendance
Uncertain event	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Mobile library.
A_2	—	Alternative: Pop-up branches.
S_1	—	State: High attendance.
S_2	—	State: Low attendance.
$q(A_1, S_1)$	50 groups	Quantity for Mobile library under High attendance.
$q(A_1, S_2)$	30 groups	Quantity for Mobile library under Low attendance.
$q(A_2, S_1)$	45 groups	Quantity for Pop-up branches under High attendance.
$q(A_2, S_2)$	35 groups	Quantity for Pop-up branches under Low attendance.
$v(S_1)$	0.7 thousand USD per group	Attendance-based funding for High attendance.
$v(S_2)$	0.4 thousand USD per group	Attendance-based funding for Low attendance.
$f(A_1)$	8 thousand USD	Program-budget allocation for Mobile library.
$f(A_2)$	11 thousand USD	Program-budget allocation for Pop-up branches.
$f(S_1)$	5 thousand USD	Additional program funding or unrecovered venue and publicity costs for High attendance.
$f(S_2)$	-2 thousand USD	Additional program funding or unrecovered venue and publicity costs for Low attendance. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i) + f(S_j).$$

Public Library Outreach: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times v(S_1)$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(S_2)$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times v(S_1)$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(S_2)$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	50×0.7 $+8+5$ $=48$	30×0.4 $+8-2$ $=18$
A_2	45×0.7 $+11+5$ $=47.5$	35×0.4 $+11-2$ $=23$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	48	18
A_2	47.5	23



Grain Terminal Schedule (C1–C2)

A grain terminal must choose a day, night, or split loading schedule. Labor contracts provide 12 thousand USD of operating income for the day schedule, 8 thousand USD for the night schedule, and 10 thousand USD for the split schedule. Open export channels earn 0.6 thousand USD per cargo lot and provide 7 thousand USD of priority-access income. Restricted channels earn 0.3 thousand USD per lot and impose 3 thousand USD of rescheduling costs. The estimates come from prior volumes, labor costs, and forecasts of export restrictions.

The day schedule is expected to handle 90 lots with open channels and 55 with restricted channels. The night schedule is expected to handle 75 and 60 lots, respectively, and the split schedule 85 and 65 lots, respectively.

The day schedule handles the most lots when channels are open, while the split schedule handles the most under restricted conditions.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Day schedule	A_2: Night schedule	A_3: Split schedule
States of nature	S_1: Open channels	S_2: Restricted channels	
Uncertain event	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Day schedule.
A_2	—	Alternative: Night schedule.
A_3	—	Alternative: Split schedule.
S_1	—	State: Open channels.
S_2	—	State: Restricted channels.
$q(A_1, S_1)$	90 cargo lots	Quantity for Day schedule under Open channels.
$q(A_1, S_2)$	55 cargo lots	Quantity for Day schedule under Restricted channels.
$q(A_2, S_1)$	75 cargo lots	Quantity for Night schedule under Open channels.
$q(A_2, S_2)$	60 cargo lots	Quantity for Night schedule under Restricted channels.
$q(A_3, S_1)$	85 cargo lots	Quantity for Split schedule under Open channels.
$q(A_3, S_2)$	65 cargo lots	Quantity for Split schedule under Restricted channels.
$v(S_1)$	0.6 thousand USD per cargo lot	Channel-dependent loading earnings for Open channels.
$v(S_2)$	0.3 thousand USD per cargo lot	Channel-dependent loading earnings for Restricted channels.
$f(A_1)$	12 thousand USD	Labor-contract operating income for Day schedule.
$f(A_2)$	8 thousand USD	Labor-contract operating income for Night schedule.
$f(A_3)$	10 thousand USD	Labor-contract operating income for Split schedule.
$f(S_1)$	7 thousand USD	Priority-access income or rescheduling costs for Open channels.
$f(S_2)$	-3 thousand USD	Priority-access income or rescheduling costs for Restricted channels. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i) + f(S_j).$$

Grain Terminal Schedule: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times v(S_1)$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times v(S_2)$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times v(S_1)$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times v(S_2)$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$
A_3	$q(A_3,S_1)\times v(S_1)$ $+f(A_3)+f(S_1)$ $=P_{ay}(A_3,S_1)$	$q(A_3,S_2)\times v(S_2)$ $+f(A_3)+f(S_2)$ $=P_{ay}(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	90×0.6 $+12+7$ $=73$	55×0.3 $+12-3$ $=25.5$
A_2	75×0.6 $+8+7$ $=60$	60×0.3 $+8-3$ $=23$
A_3	85×0.6 $+10+7$ $=68$	65×0.3 $+10-3$ $=26.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	73	25.5
A_2	60	23
A_3	68	26.5



6 Complete Four-Component Models

Problems:

6.1. Urban Food Hub (C1–C2)
6.1. Regional Rail Cargo (C1–C2)

6.1 Alternative-Based Variable Plus State-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

Problems:

6.1. Urban Food Hub (C1–C2)
6.1. Regional Rail Cargo (C1–C2)

Urban Food Hub (C1–C2)

A city food hub must choose electric or manual sorting for produce crates. The electric system's service contract provides 9 thousand USD of income and earnings of 0.6 thousand USD per crate. Manual sorting receives 14 thousand USD under its contract and earns 0.4 thousand USD per crate. Strong demand adds 0.2 thousand USD to the earnings

from each crate and brings 6 thousand USD of distributor incentives. Soft demand adds 0.1 thousand USD per crate, while unused-capacity costs reduce profit by 3 thousand USD. These estimates come from distribution forecasts, labor costs, equipment expenses, and recent demand.

The hub expects electric sorting to process 70 crates under strong demand and 45 under soft demand. Manual sorting is expected to process 80 and 55 crates, respectively.

Electric sorting earns more per crate before demand-related income, while manual sorting receives more contract income and processes more crates under both demand conditions.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Electric sorting	A_2**:** Manual sorting
States of nature	S_1**:** Strong demand	S_2**:** Soft demand
Uncertain event	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Electric sorting.
A_2	—	Alternative: Manual sorting.
S_1	—	State: Strong demand.
S_2	—	State: Soft demand.
$q(A_1, S_1)$	70 produce crates	Quantity for Electric sorting under Strong demand.
$q(A_1, S_2)$	45 produce crates	Quantity for Electric sorting under Soft demand.
$q(A_2, S_1)$	80 produce crates	Quantity for Manual sorting under Strong demand.
$q(A_2, S_2)$	55 produce crates	Quantity for Manual sorting under Soft demand.
$v(A_1)$	0.6 thousand USD per produce crate	Sorting earnings for Electric sorting.
$v(A_2)$	0.4 thousand USD per produce crate	Sorting earnings for Manual sorting.
$v(S_1)$	0.2 thousand USD per produce crate	Demand-related earnings for Strong demand.
$v(S_2)$	0.1 thousand USD per produce crate	Demand-related earnings for Soft demand.
$f(A_1)$	9 thousand USD	Service-contract income for Electric sorting.
$f(A_2)$	14 thousand USD	Service-contract income for Manual sorting.
$f(S_1)$	6 thousand USD	Distributor incentives or unused-capacity costs for Strong demand.
$f(S_2)$	-3 thousand USD	Distributor incentives or unused-capacity costs for Soft demand. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i) + f(S_j).$$

Urban Food Hub: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$70\times$ $[0.6+0.2]$ $+9+6$ $=71$	$45\times$ $[0.6+0.1]$ $+9-3$ $=37.5$
A_2	$80\times$ $[0.4+0.2]$ $+14+6$ $=68$	$55\times$ $[0.4+0.1]$ $+14-3$ $=38.5$

Final payoff table (thousand USD).

Alternative	S_1	S_2
A_1	71	37.5
A_2	68	38.5



Regional Rail Cargo (C1–C2)

A rail operator must choose an express or flexible service for moving freight lots. The express contract provides 10 thousand USD of service income and earnings of 0.8 thousand USD per lot. The flexible contract provides 15 thousand USD and earnings of 0.6 thousand USD per lot. Clear conditions add 0.2 thousand USD to the earnings from each lot and produce 5 thousand USD of on-time-delivery bonuses. Busy conditions add 0.1 thousand USD per lot and bring no additional bonus or cost. Disruptions reduce earnings by 0.1 thousand USD for every lot and impose a further 7 thousand USD of rerouting costs. The estimates come from past volumes, operating costs, and network simulations.

Volume forecasts show the express service moving 85 lots under clear conditions, 65 under busy conditions, and 40 under disrupted conditions. The flexible service is expected to move 75, 70, and 50 lots under those respective conditions.

Express service earns more for each lot before network-related income or costs, while flexible service maintains higher forecast volumes under busy and disrupted conditions.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Express service	A_2: Flexible service	
States of nature	S_1: Clear network	S_2: Busy network	S_3: Disrupted network
Uncertain event	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		



C2 - Building the payoff table.
Symbols, values, and meanings.

Symbol	Value	Meaning in this problem
A_1	—	Alternative: Express service.
A_2	—	Alternative: Flexible service.
S_1	—	State: Clear network.
S_2	—	State: Busy network.
S_3	—	State: Disrupted network.
$q(A_1, S_1)$	85 freight lots	Quantity for Express service under Clear network.
$q(A_1, S_2)$	65 freight lots	Quantity for Express service under Busy network.
$q(A_1, S_3)$	40 freight lots	Quantity for Express service under Disrupted network.
$q(A_2, S_1)$	75 freight lots	Quantity for Flexible service under Clear network.
$q(A_2, S_2)$	70 freight lots	Quantity for Flexible service under Busy network.
$q(A_2, S_3)$	50 freight lots	Quantity for Flexible service under Disrupted network.
$v(A_1)$	0.8 thousand USD per freight lot	Service earnings for Express service.
$v(A_2)$	0.6 thousand USD per freight lot	Service earnings for Flexible service.
$v(S_1)$	0.2 thousand USD per freight lot	Network-related earnings or reductions for Clear network.
$v(S_2)$	0.1 thousand USD per freight lot	Network-related earnings or reductions for Busy network.
$v(S_3)$	-0.1 thousand USD per freight lot	Network-related earnings or reductions for Disrupted network. Negative: reduces the payoff.
$f(A_1)$	10 thousand USD	Service-contract income for Express service.
$f(A_2)$	15 thousand USD	Service-contract income for Flexible service.
$f(S_1)$	5 thousand USD	Delivery bonuses or rerouting costs for Clear network.
$f(S_2)$	0 thousand USD	Delivery bonuses or rerouting costs for Busy network.
$f(S_3)$	-7 thousand USD	Delivery bonuses or rerouting costs for Disrupted network. Negative: reduces the payoff.

The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i) + f(S_j).$$

Regional Rail Cargo: payoff calculations.
All calculation cells and final payoffs are in thousand USD. The indices refer to the symbols table above.

Payoff calculations

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$q(A_1,S_1)\times$ $[v(A_1)+v(S_1)]$ $+f(A_1)+f(S_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v(A_1)+v(S_2)]$ $+f(A_1)+f(S_2)$ $=P_{ay}(A_1,S_2)$	$q(A_1,S_3)\times$ $[v(A_1)+v(S_3)]$ $+f(A_1)+f(S_3)$ $=P_{ay}(A_1,S_3)$
A_2	$q(A_2,S_1)\times$ $[v(A_2)+v(S_1)]$ $+f(A_2)+f(S_1)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v(A_2)+v(S_2)]$ $+f(A_2)+f(S_2)$ $=P_{ay}(A_2,S_2)$	$q(A_2,S_3)\times$ $[v(A_2)+v(S_3)]$ $+f(A_2)+f(S_3)$ $=P_{ay}(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$85\times$ $[0.8+0.2]$ $+10+5$ $=100$	$65\times$ $[0.8+0.1]$ $+10+0$ $=68.5$	$40\times$ $[0.8-0.1]$ $+10-7$ $=31$
A_2	$75\times$ $[0.6+0.2]$ $+15+5$ $=80$	$70\times$ $[0.6+0.1]$ $+15+0$ $=64$	$50\times$ $[0.6-0.1]$ $+15-7$ $=33$

Final payoff table (thousand USD).

Alternative	S_1	S_2	S_3
A_1	100	68.5	31
A_2	80	64	33